SALISHAN

BY

DOUGLAS QUIGG

RESEARCH AND THEORY
WHOSE WORDS THESE ARE, AND HOW WRONG THE CORPUS COULD BE

ORIOR
2026

© 2026 Douglas Quigg
AGPL-3.0-or-later OR LicenseRef-Commercial OR LicenseRef-Educational

For Megan, my loving wife, the only person that really understands what I mean when I speak.

For Autumn and Opal, without you this thought would have never occurred.

To everyone else, I wish you well.

ABSTRACT

The corpus everything in this work is measured against is Salishan speech, written down. This research paper says whose words it holds, how wrong it could be, and how that was measured.

The corpus is a glossed word list: each form is taken out of a published paper and paired with its meaning, and the pairs are the research's own data. Every table opens with the person who spoke, before the linguist who published and before anyone who transcribed the paper into a table. Twenty-five tables carry 16,822 rows. Sixteen name their speakers. Nine cite a published source and name no speaker, and those say so instead of guessing. A linguist wrote the paper and a person transcribed it, and neither of those is whose language it is.

How wrong the corpus could be is stated as an upper bound, never as a measurement of the rate. Three channels fail for different reasons: a form written that the paper does not hold, a token the paper prints that no row holds, and a token the corpus lost on the way out of an automatic extractor. The checks observe 163 failures in 20,261 trials, 28 in 7,440, and 4 in 9,430. Taken as independent the joint bound is 1.28×10^{-8} per line. The derivation says where that independence is thin, and the largest unmodelled term is that one person transcribed every table.

The method finds a border it is never displayed. Mellesmoen and Kye label every form they cite as northern or southern Lushootseed, a fact a linguist published. Given the forms and never the labels, holding the concept fixed through the word web and flattening the pooled run counts to max entropy, the term that carries the border comes back as the stressed schwa, southern, at 45 against 6 over the same concepts, matched by 1 of 200 random borders.

Two papers here are set in a font with no Unicode map. Their extracted text is the font's encoding and not the page. Their page texts are drafted rule by rule and remain drafts: a rule that guesses at labialization is named as guessing. Five more papers hold none of their own orthography in their text layer, and none of them is counted for or against the corpus.

These methods read a language and can put one back. Near the center of a subject a regeneration is a copy, and far from it the output is obvious nonsense. Between them is a narrow band where the output is coherent and may already not be the language, and nothing here marks which side of it a result fell on. That question belongs to a native speaker, and no amount of measurement turns it into one an algorithm can answer.

CONTENTS

1 WHOSE WORDS THESE ARE . . . 1
- 1.0.1 Lushootseed . . . 1
- 1.0.2 Lushootseed . . . 1
- 1.0.3 St'át'imcets . . . 1
- 1.0.4 St'át'imcets . . . 2
- 1.0.5 Nuxalk . . . 2
- 1.0.6 nɬeʔkepmxcín . . . 2
- 1.0.7 nɬeʔkepmxcín . . . 2
- 1.0.8 Mainland Comox (ayajuthem) . . . 2
- 1.0.9 nɬeʔkepmxcín . . . 3
- 1.0.10 Nsyilxcən . . . 3
- 1.0.11 Nsyilxcən . . . 3
- 1.0.12 Lushootseed . . . 3
- 1.0.13 nɬeʔkepmxcín and Secwepemctsín . . . 4
- 1.0.14 eighteen Central Salish languages . . . 4
- 1.0.15 Nuxalk . . . 4
- 1.0.16 Nsyilxcən . . . 4
- 1.0.17 Twana . . . 4
- 1.0.18 Nuxalk . . . 5
- 1.0.19 nɬeʔkepmxcín . . . 5
- 1.0.20 St'át'imcets . . . 5
- 1.0.21 Tillamook . . . 5
- 1.0.22 nɬeʔkepmxcín . . . 6
- 1.0.23 Snohomish . . . 6
- 1.0.24 Twana . . . 6
- 1.0.25 Columbian . . . 7

2 THE CORPUS UNDER THE INSTRUMENT . . . 9
- 2.1 The permutation-null method on Salishan . . . 9
  - 2.1.1 The idea . . . 9
  - 2.1.2 The procedure . . . 9
  - 2.1.3 The components . . . 9
  - 2.1.4 Results . . . 10
  - 2.1.5 What it cannot do . . . 11

3 HOW WRONG IT COULD BE . . . 13
- 3.0.1 1. What is being estimated . . . 13

3.0.2 2. What the checks have seen . . . 13
3.0.3 3. The channels . . . 14
3.0.4 4. What the number rests on . . . 16
3.0.5 5. Automatic extractors against their tables . . . 16
3.0.6 6. The word web . . . 17
3.0.7 7. The dialect border . . . 17

4 REPRODUCING WHAT THE LINGUISTS CLAIMED . . . 19
4.1 Nater on Bella Coola . . . 19
4.1.1 Why this is a different type of test . . . 19
4.1.2 What had to be thrown out first . . . 19
4.1.3 The reproduction . . . 20
4.1.4 Whether 0.136 is anything . . . 20
4.1.5 The claim about nuclei . . . 21
4.1.6 What this cannot do . . . 21

5 FROM THE PAGE TO A DISTRIBUTION OVER SOUNDS . . . 23
5.1 Three stages, and they are not interchangeable . . . 23
5.2 Stage zero, the page . . . 23
5.3 Stage one, characters . . . 24
5.4 Stage two, segments . . . 24
5.5 Stage three, sounds . . . 24
5.5.1 Where the correspondences come from . . . 25
5.5.2 Two collisions the table must avoid . . . 25
5.5.3 What the mapping does not claim . . . 25
5.6 What a distribution reads at each stage . . . 25
5.7 Limitations . . . 25

6 THE RECORDINGS UNDER THE INSTRUMENT . . . 27
6.1 What the prosody field holds . . . 27
6.1.1 The question . . . 27
6.1.2 The recordings . . . 27
6.1.3 The answer, through the necessary condition . . . 28
6.1.4 What that does not settle . . . 28
6.1.5 Who spoke is read from the paper . . . 28
6.2 The segment field is not comparable between recordings . . . 29
6.2.1 What the representation does . . . 29
6.2.2 A comparison that does not isolate the rotation . . . 29
6.2.3 What isolates it . . . 29
6.2.4 Why the prosody field escapes it . . . 30
6.3 The waveform itself, with nothing thrown away . . . 30

6.3.1 Why the representation cannot answer this . . . 30
6.3.2 A linear slice reads the silence . . . 30
6.3.3 Where the arrangement sits . . . 31
6.3.4 There is no unit scale in it, and that absence is the result . . . 31
6.3.5 What that settles about reading speech back . . . 31
6.4 The join between the web and the sound . . . 32
6.4.1 The missing stage . . . 32
6.4.2 The supervision is printed on the page . . . 32
6.4.3 A segment is a grapheme cluster and not a character . . . 32
6.4.4 Placing a segment inside its sentence, and why the crude answer is allowed . . . 33
6.4.5 What the pooled sound gives . . . 33
6.4.6 What it needs next, which the measurement names . . . 33

7 MATERIALS, PROCEDURE AND AVAILABILITY . . . 35
7.1 Source texts . . . 35
7.2 When the text is not the page . . . 35
7.3 Procedure . . . 35
7.4 Shared definitions . . . 36
7.5 The sound representation . . . 36
7.6 Data and code availability . . . 37

8 SOURCES . . . 39
8.1 Speakers . . . 39
8.2 Papers with an automatic extractor . . . 39
8.2.1 Surface and underlying forms . . . 40
8.2.2 An etymological lexicon . . . 40
8.2.3 Twana . . . 41
8.2.4 Elicited roots . . . 42
8.2.5 Words with no vowel . . . 42
8.2.6 A comparative reconstruction . . . 42
8.3 Hand extraction . . . 42
8.4 Papers whose extracted text is not the page . . . 44
8.4.1 A scanned typescript is a worse case than any font . . . 46
8.4.2 The damage is a mutation distribution . . . 46
8.4.3 The per-run test reads the damage a blanket repair cannot . . . 48
8.5 What the gold standard corpus measures . . . 49
8.6 What a gold standard corpus is for . . . 50
8.7 The binary sound representation . . . 52
8.8 Audio sources . . . 52
8.9 Unread papers . . . 53

8.10 Verification sources 53
8.11 Community sources 54
8.12 Archives 54

BIBLIOGRAPHY 55

1

WHOSE WORDS THESE ARE

These languages belong to the people who speak them. None of this work exists without them. Everything else in this research is measured against the tables below, and the tables are their words, written down.

Each entry opens with who spoke, because that is whose language it is. A linguist wrote the paper and a person read the paper into a table, and neither of those is whose language it is. Where a paper cites a published dictionary and never says who spoke, the entry says so, and the linguist does not go in the speaker column.

Conditions the speakers set are recorded with them below and hold wherever this corpus is used.

1.0.1 *Lushootseed*

- **Martha Lamont, Northern dialect**
- **Annie Jack, Southern dialect**

[25], 467 rows.

Both recorded by Leon Metcalf in the 1950s. The only paper here that labels every form by dialect, and the border test is scored against it.

1.0.2 *Lushootseed*

- **Susie Sampson Peter, Upper Skagit**
- **Martha LaMont, Tulalip-Skagit**

[13], 60 rows.

Vi taqʷšəblu Hilbert wrote the paper; the twenty-one examples were said by her aunt Susie Sampson Peter and by Martha LaMont, recorded by Leon Metcalf between 1950 and 1958 and by Thom Hess in 1963. Hilbert is the author of the paper and not its speaker.

1.0.3 *St'át'imcets*

- **Ḱweswapáẇ (Linda Redan), Qayqáyten**

- **Sam Mitchell, in van Eijk and Williams 1981**

[24], 104 rows.
Ẋweswapáẇ told the story over Zoom on 31 October 2025, three minutes twenty-eight seconds, and the audio and video are retained by her. Sam Mitchell is the speaker of the earlier text the paper cites.

1.0.4 *St'át'imcets*

- **Qwa7yán'ak (Carl Alexander), Nxwísten**

[2], 691 rows.
Recorded at Nxwísten on 7 July 2025, just over half an hour.

1.0.5 *Nuxalk*

- **Dr. Margaret Siwallace**

[27], 285 rows.
Recorded about 1975, published 2015.

1.0.6 *nɬeʔkepmxcín*

- **wlwlmelst (Maurice Michell), Southern yutémkt dialect**

[18], 226 rows.
He shares his four stories freely for people connecting with the language. They came from his mother nxwelinek and his grandmother ʔústko.

1.0.7 *nɬeʔkepmxcín*

- **Kʷəɬtəzétkʷu (Bernice Garcia), ċəɬétkʷu (Coldwater)**

[6], 371 rows.
She asks it be acknowledged she is a Kamloops Indian Residential School speaker re-learning her language.

1.0.8 *Mainland Comox (ayajuthem)*

- **Mary George, Sliammon**
- **Noel George Harry**
- **Tommy Paul**

[15], 1661 rows.

Recorded 1969 to 1980. The two checks of its extraction repair it differently, and their results for it are not directly comparable.

1.0.9 *nɬeʔkepmxcín*

- **Bev Phillips, Lytton First Nation (x̌ʷq̓əmcín)**

[9], 406 rows.

Her own recorded reading of the story is stored with the corpus, and it checks the extraction as well as sourcing it. The two checks of its extraction close spaces differently.

1.0.10 *Nsyilxcən*

- **George Lezard, Penticton Indian Reserve**
- **Nellie Guitterez, Upper Nicola Indian Band**
- **Kiláwnaʔ (Andrew McGinnis), Penticton Indian Reserve**

[22], 1417 rows.

George Lezard recorded 1966 by Randy Bouchard, transcribed by Larry Pierre 1970, updated by permission of Arnie Baptiste, his son. Nellie Guitterez recorded 1978 or 1979 by Yvonne Hébert, reprinted by permission of Lynne Jorgesen, her great-granddaughter.

1.0.11 *Nsyilxcən*

- **Lottie Lindley, Upper Nicola**

[19], 653 rows.

1.0.12 *Lushootseed*

- The paper names no speaker. Its forms are cited from a published source.

[14], 74 rows.

A 1975 typescript, scanned, and its text is OCR of the scan. The paper names no speaker for its examples. The OCR carries none of the orthography: zero schwas, zero raised w, zero barred l, zero wedges, against 169, 77, 41 and 33 in the hand extraction. It writes taqWsablu for taqʷšəblu and slahal for sləhal. All 200 disagreements in which the extraction holds a form the hand extraction does not come from that, and they are faults of the source text and not of the transcriber. Until a page text is drafted for it, as for the two Lyon papers, a check against it measures nothing.

1.0.13 *nɬeʔkepmxcín and Secwepemctsín*

- **Charley Alexis Mayoos**
- **William Celestin**

[31], 195 rows.
Their texts are written in Chinuk pipa. Texts 3 to 6 are Chinook Jargon, which is a pidgin and is not Salish, and the speaker column retains those out.

1.0.14 *eighteen Central Salish languages*

- The paper names no speaker. Its forms are cited from a published source.

[33], 772 rows.
Every form is cited from a published dictionary of one of eighteen languages. there is nobody this corpus is of. The speaker column carries the language instead, and the gold standard corpus for this paper is retained per language.

1.0.15 *Nuxalk*

- The paper names no speaker. Its forms are cited from a published source.

[28], 671 rows.
Nater's own records, from his 1990 dictionary and 1984 grammar. No speaker is named. Six entries and two tables are Heiltsuk, Oowekyala, Kwak̓wala and Haisla, which are North Wakashan and not Salish at all.

1.0.16 *Nsyilxcən*

- **ɬk̓mxnalqs (Delphine Derrickson-Armstrong), stq̓aʔtkʷɬniẇt**
- **c̓əskʕákənaʔ (Dave Michele), stq̓aʔtkʷɬniẇt**

[23], 537 rows.
Elicited from both speakers. Max cells of its two tables are starred, which is a form the linguist built and the speakers rejected, and those are excluded.

1.0.17 *Twana*

- The paper names no speaker. Its forms are cited from a published source.

[16], 358 rows.
Every Twana form is Drachman's, out of a 1969 dissertation the paper calls the only reliable reference in existence for this. No speaker is named. Footnote 9's four Moses-Columbian forms are the only place its extraction repeats a combining mark, and those are corrected from the page.

1.0.18 *Nuxalk*

- The paper names no speaker. Its forms are cited from a published source.

[26], 5767 rows.
1407 numbered entries out of Nater's own 1990 dictionary. No speaker is named.

1.0.19 *nɬeʔkepmxcín*

- **Bev Phillips**
- **ċúʔsinek (Marty Aspinall)**
- **kʷałtèzetkʷ (Bernice Garcia)**

[10], 494 rows.
The forms are cited from Thompson and Thompson's grammar and dictionary. These three are the speakers the paper thanks, two examples are Bev Phillips reading her own story, and kʷałtèzetkʷ introduces herself in the acknowledgement.

1.0.20 *St'át'imcets*

- **Qwa7yán'ak (Carl Alexander), Nxwísten**

[4], 441 rows.
Its data has three sources: van Eijk's dictionary, Davis et al. in preparation, and elicitation with Carl Alexander. It labels forms (U) and (L) for Upper and Lower St'át'imcets, the second external dialect label in the archive.

1.0.21 *Tillamook*

- The paper names no speaker. Its forms are cited from a published source.

[11], 154 rows.
Eric P. Hamp, Another Look at Tillamook Phonology, ICSNL 2. It cites no words. its language content is four consonant inventories set side by side and two feature matrices. No speaker is named, and the paper works throughout from Thompson and Thompson, Reichard, Kinkade, Drachman and Edel. What it does record is that a Tillamook speaker was living in 1967 and it does not say who: the urgency of the matter is put as being that there is yet a surviving speaker available for possible re-check, and page 2 grants one sense of its claim only on the assumption that a last remaining speaker is typical of a community. Both sentences are in the table. The Twana chart is used, in the page's own words, without his permission, meaning Drachman's. The dot under the uvular fricatives prints solid in some cells of these charts and as an open ring in others, and the two positions swap between the Tillamook and Twana charts. It is one mark and the variation is the typewriter.

1.0.22 *nɬeʔkepmxcín*

- **Bev Phillips, Lytton (x̌ʔq̓əmcín) dialect**

[7], 178 rows.

Katherine Givens and Brent Hall, The Moon and the Birchbark Canoe (ɬ máʕxetn pe ɬ qʷɬinéwɬ), ICSNL 58. Seven pages, read one at a time off page renders. Page 1: the story was recounted in Nɬeʔkepmxcín by Bev Phillips, a native speaker of the Lytton dialect, who also helped with the translation, and Givens and Hall transcribed and glossed it. She is quoted on the page choosing the story and saying creation stories are not just stories to us, and the paper's first footnote thanks her for entrusting it to them. Of the three recordings, only this one has a file named for the transcribers and not the speaker, and read from that name alone it suggests a second speaker. Footnote 4 defines (VG), a volunteered gloss, as a translated sentence BP offered. The parenthesis at the right margin is the only thing separating her English from the authors', and examples 12, 13 and 14 lack it. Its section 4 repeats every sentence of section 2 in morpheme-broken form, which gave a second independent reading of every word and turned up three places where the two tiers disagree. Footnotes 2, 6, 7 and 8 each declare a mark or a parsing the authors reached by ear or by inference, which is more than any other paper here states about its own readings.

1.0.23 *Snohomish*

- The paper names no speaker. Its forms are cited from a published source.

[12], 248 rows.

Thom Hess, The Morph /-(ə)b/ in Snohomish, ICSNL 2. The hand extraction is checked against the paper. No speaker is named. The forms are Hess's Snohomish data. Marked characters, page 3, by eye and inside one table: xáyəb 'laugh' and xʷúyub 'sell' print a bare x where x̌áɬəb 'salty' prints a barred x body with a glottalization mark, ƭádəb 'bitter' prints the same t as the English word taste on the first line of that page with a glottalization mark added, and ƌáx̌əb 'cloud' carries a hook at the top of its d and a bar across the foot, one composite, where the d of ƭádəb one column away is bare. Below sixteen times magnification either half of that composite drops out. Glyphs in the printed text unclear. Hilbert and Hess (1975) set the same character in dəxʷgʷəx̌ələds, and Nater's Coast Salish *x̌aɬ 'bitter, salt' matches x̌áɬəb segment for segment.

1.0.24 *Twana*

- The paper names no speaker. Its forms are cited from a published source.

[5], 276 rows.

William W. Elmendorf, Word Tabu and Change Rates, ICSNL 2. Thirteen languages are cited in it and Twana is its subject. That is the language named here; every row carries its own language in the speaker column. No speaker is named anywhere: the forms come from Boas and Haeberlin 1927, Krueger 1967, McIlwraith 1948, Walters 1938 and Ray 1932, from Warren Snyder's Suquamish list and Wayne Suttles' Squamish field notes, and from Elmendorf's own field

data. This typescript writes its uvular as x under a dot where Hilbert and Hess (1975) write it under a caron, and Hess (1967) agrees with it, which is two papers of one conference against one of a later year. One mark is unresolved and marked so in the table: a short raised stroke over the s of the Columbia ska'u on page 7, which is not the wedge the Upper Chehalis sča'u carries two words earlier.

1.0.25 *Columbian*

- The paper names no speaker. Its forms are cited from a published source.

[17], 316 rows.

M. Dale Kinkade, Deictics in Columbian: A Work Paper, ICSNL 2. Twelve pages, read one at a time off page renders. Columbian is what it is about; Kalispel comes from Vogt 1940, Coeur d'Alene from Reichard 1938, and three Colville forms from an unnamed speaker who also knew Columbian. Every row carries its own language in the speaker column. No speaker is named anywhere in the paper: it says my Cm informants and says no more than that. The forms in this table were said by people it does not identify. Its typewriter has three raised marks and the table's symbol notes turn on telling them apart, a comma with a thick head and a curling tail, a V wedge, and a straight acute. čéṅ on page 10 carries all three in one word and is the control. The wedge appears only in the Kalispel and Coeur d'Alene forms, ten times for ten, and never in the Columbian ones, whose seven caron readings are assigned and are all x̌ or the č of čiílx. Whether that split is about the language or about page order is not decidable here, because the wedge first appears on page 8 and every Columbian form was typed on pages 1 to 7. Elmendorf (1967), from the same conference, distinguishes a wedge from a short raised stroke in its own table. the distinction is one these typescripts can carry.

25 tables, 16822 rows read by hand.

2

THE CORPUS UNDER THE INSTRUMENT

2.1 THE PERMUTATION-NULL METHOD ON SALISHAN

This chapter applies the method of the delta null research paper [29] to the Salishan papers: it finds the language in a paper, and decides what may join a gold standard corpus.

2.1.1 *The idea*

English is the target. The object of study is what English does not account for.

That turns the problem around, onto the half that is well resourced. Nobody has to know Nsyilxcən or Lushootseed to find it in a paper. English has to be known, and the residual text is the language, a gloss, or a page of font damage.

2.1.2 *The procedure*

1. Build a gold standard corpus for each language from the hand extractions.
2. Build an English reference corpus from the translations the hand extractions mark.
3. Screen each paper against the reference corpus, and retain what English does not account for.
4. Measure what was retained against each language's profile, and admit it only where the measurement says it can be read (Section 3).

2.1.3 *The components*

The measurement reduces a text to its byte-pair histogram, a bigram distribution [32], and compares two texts by the total variation distance between their histograms. A language profile is the histogram of one language's gold standard corpus, in the sense of *n*-gram language identification [3]. Each profile carries a split-half distance, the distance between two halves of its own corpus, and a comparison between corpora has to beat it to be read. The measurement has no tuned parameter and no term for any one language. Every distance, verdict and median below is an exact rational, computed in integer arithmetic on the engine's record machine (`gold_readings.cu` in the orior repository).

Around it sit three other components: an automatic extractor written for each of 18 papers, with that paper's repairs and checks; the screening applied to papers with no extractor, together with the dialect border test; and the word web, which joins the forms of every hand extraction by concept, shape and context.

2.1.4 *Results*

These results are over 25 hand extractions. Of these, 18 have an automatic extractor, and every language token of 15 of those is accounted for:

- 25 of 25 papers have a hand extraction;
- 15 of 18 papers have every language token accounted for;
- 2303 lines of gold standard target language, across 6 languages.

The 7 with no extractor, Hilbert and Hess [14] among them, are read by hand, and they are retained ungraded and not counted. The 6 languages are the ones with a gold standard corpus.

There are 7 profiles: 6 languages and English. Of the 21 pairs, 20 are readable, at distances 0.512 to 0.834 against split-half distances of 0.148 to 0.641. The pair that does not read is Lushootseed against Nsyilxcən, 0.622 apart with the Lushootseed profile's own split-half distance at 0.641.

That split-half distance is not the resolution of the estimator at 219 lines. Cutting a corpus at its midpoint measures the order its papers are in: a corpus is its papers concatenated, the cut lands on a paper boundary, and it reports the distance between whichever papers it separates. Lushootseed is two papers in two dialects and two orthographies, and the midpoint puts one on each side. Sampling the halves by alternating puts a mixture on both sides, and the split-half distance falls to 0.300. Every corpus moves the same way, by 1.32 to 3.76 times, and the midpoint figure is the larger every time. A resolution cannot rise when its sample grows, and the midpoint figure does. The split-half distances are therefore sampled by alternating.

Read at the alternating split-half distances, all 15 pairs among the 6 languages separate, Lushootseed against Nsyilxcən included, and 1 of the 15 change verdict. The 21 above counts the same 6 languages plus English.

The papers are a separate question, and the profiles' split-half distances do not decide it. Each paper is judged against its own split-half distance. The 9 papers whose readers made the profiles are left out, since each would be judged against its own text. None of the 277 papers naming a language the profiles cover is readable from the lines its English screen leaves. Each holds at least 2000 byte pairs, and each fails for one reason: the gap between the nearest profile and the runner-up has a median of 0.0470, against a median paper split-half distance of 0.2012. The gap is 4 times inside the noise.

Read instead from their tool extractions, the lines the screening below places nearer the pure corpus than English, the same papers give a different answer. Of the 277, 8 are readable, and on all 8 the distributions agree with what the paper's prose says the language is. Cut to those lines, 168 fall below 2000 byte pairs, and the other 101 fail as the screened papers do, with a median gap of 0.0438 against a median split-half distance of 0.3692.

Screened from the papers with no extractor:

- 979 papers screened;
- 77828 lines nearer the language, and 171956 lines of residual text for a person to review;
- 542 of the 979 carry the language their own front matter names.

Admission is decided on each corpus's own growth curve:

Language	Gold standard	Candidates	Admitted	Split-half before	after	Support after
Comox	426	6756	0	0.1956	0.1956	407
Lushootseed	219	2779	2779	0.6411	0.4424	3570
Nsyilxcən	738	7362	40	0.1561	0.1823	898
Nuxalk	98	3927	0	0.2536	0.2536	346
St'át'imcets	371	5323	3040	0.2768	0.3440	3302
nɬeʔkepmxcín	451	4908	4680	0.3098	0.3833	2707

Comox and Nuxalk refused all of theirs: adding the whole set to Comox takes its support from 407 cells to 4651 and its split-half distance from 0.196 to 0.283, a second distribution arriving and not more of the first.

Lushootseed took all 2779, and its split-half distance fell from 0.6411 to 0.4424. The admission rule is only a ceiling, and a corpus sitting above the band it belongs in admits whatever pulls it down. Here 0.6411 is two dialects on opposite sides of the midpoint cut, and 2779 candidates added between them dilute that split without adding anything to the profile. A falling split-half distance can be a boundary being blurred and not a corpus improving.

Candidates are retained for 18 languages that have no hand-read corpus to grow: Halkomelem 4132, Secwepemctsín 2939, Upper Chehalis 1777, Nxaʔamxcín 1089, Squamish 999, Nooksack 655, Straits 599, Montana Salish 563, Coeur d'Alene 369, Cowlitz 364, Tillamook 334, Twana 254, Klallam 247, Sechelt 142, Spokane 126, Kalispel 56, Tsamosan 56, Quinault 3.

2.1.5 *What it cannot do*

Support is still climbing on every one of these corpora. None has seen its own alphabet. A candidate that looks nothing like a member is not thereby wrong, and resembling what is already there is the wrong test.

The screening finds candidates. It does not read them. The corpus is still hand extracted and verified in three passes, for line and for notation, and the screening says where to look.

Growing the Lushootseed profile does not reach the 277 papers. The 277 that are large enough to ask fail because a paper's residual text sits almost the same distance from all 6 language profiles: the median gap between the nearest and the runner-up is 0.0470, and the median paper split-half distance is 0.2012. Nothing about the profiles' own sizes moves those two numbers past each other.

What the 277 need is a measure that separates the profiles as seen from a mixed residual, which the byte-pair histogram does not. The dialect border section of the chapter *How wrong it could be* gives the first one that does: holding a concept fixed with the word web's gloss edge, flattening the pooled counts to max entropy, and testing one run at a time. That recovers a published dialect border inside Lushootseed at a corpus size where the whole-distribution comparison needs 2.9 times more text. It has not been applied to the 277.

Two dialects pooled into one profile are still worth splitting, because pooling inflated the split-half distance from 0.300 to 0.641, and it is the same defect in every corpus that holds more than one paper. Splitting them corrects the profiles; it is not a route to the 277.

3

HOW WRONG IT COULD BE

3.0.1 *1. What is being estimated*

For one line of the gold standard corpus, the probability that it is not what it claims to be: not the target language, or not what the paper printed. Every number below is an upper bound on that, never a measurement of it, because zero failures in a sample is not a rate of zero.

With no failures in N independent trials the true rate is under 3/N with 95 percent confidence. That bound is the rule of three, and it turns a clean check into a number. Where a check did see a failure, the observed rate is reported instead, because it is larger than any bound would be.

3.0.2 *2. What the checks have seen*

A paper contributes trials only where its extracted text is what the page prints. The others are checked against a damaged source. Their disagreements measure the source and not the table. The references chapter names each of them.

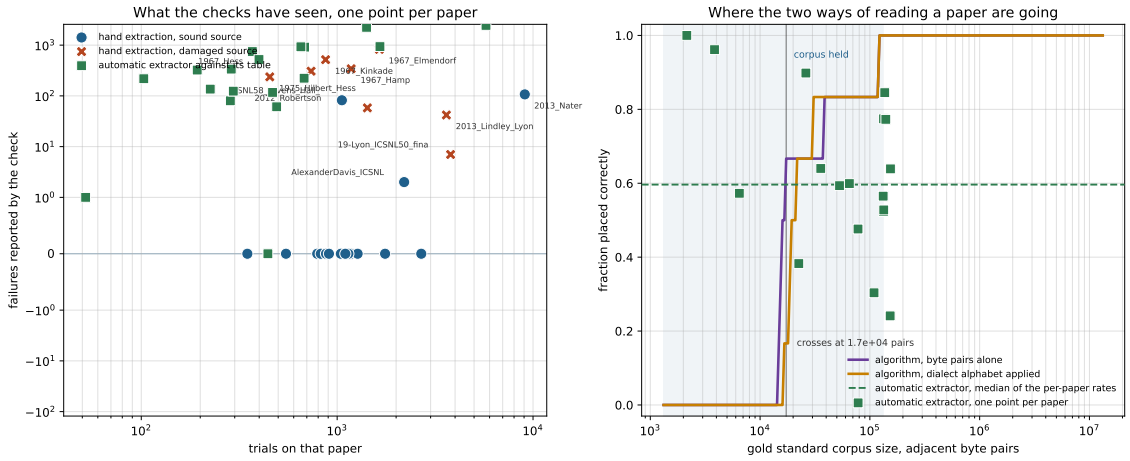


Figure 3.1: Hand extraction against its paper

Why the left panel is here. The bound in Section 3 is a single number and a single number cannot show whether it came from one clean paper or from 17. The panel puts every paper on the picture so the shape of the evidence is visible: how much each one contributed, and which ones failed anything. It shows which papers the number rests on.

What to read off it. Every paper is one point, trials against failures. The 14 on the zero line are the evidence. The 11 above it are checked against a text that is not what their page prints. The squares are a separate measurement, the automatic extractors, and they sit high: an extractor written for one paper gets a great deal wrong. Section 5 gives those per paper, the only form they mean anything in.

Why the right panel is here. The extraction has a lifetime. What carries it? Manual transcription reads a paper at a fixed accuracy however large the corpus gets. That arm is flat. The algorithm's accuracy is a function of corpus size. That arm climbs. Whether and where they cross decides whether the corpus is worth growing for its own sake, and the panel is what answers it.

What to read off it. The algorithm arm is the fraction of dialects whose distance from the other corpora pooled into one reference corpus clears twice their resolution. The arm starts at zero, because a small corpus resolves nothing, and it crosses the manual arm at 1.73e+04 byte pairs, which is smaller than the corpus already stored. Both arms are drawn: byte pairs alone, and byte pairs with the dialect's own alphabet applied first. Setting an algorithm given nothing beside a person given a page is not a fair comparison, and the alphabet is the first part of what the extraction already knows that it can be passed.

Paper	Rows	Forms checked	Not in paper	Tokens checked	No row holds	Counts
[25]	467	437	0	354	0	yes
[13]	60	445	0	103	0	yes
[24]	104	225	0	123	0	yes
[2]	691	1916	2	282	0	yes
[27]	285	579	0	313	0	yes
[18]	226	829	0	349	0	yes
[6]	371	784	0	488	0	yes
[15]	1661	2131	0	557	0	yes
[9]	406	1118	0	639	0	yes
[22]	1417	2788	5	1007	2	no
[19]	653	2424	18	1187	24	no
[14]	74	428	212	25	25	no
[31]	195	1324	38	102	20	no
[33]	772	763	0	378	0	yes
[28]	671	762	0	282	0	yes
[23]	537	666	0	435	0	yes
[16]	358	528	0	297	0	yes
[26]	5767	7191	81	1885	26	yes
[10]	494	508	0	369	0	yes
[4]	441	483	0	425	0	yes
[11]	154	1178	341	1	1	no
[7]	178	896	80	161	2	yes
[12]	248	720	291	16	16	no
[5]	276	1445	631	197	196	no
[17]	316	872	516	1	1	no

17 of the 25 hand extractions are checked against a sound source. Together they put 20261 distinct written forms and 7440 language tokens through the two directions.

3.0.3 *3. The channels*

A wrong line arrives through one of three channels. They are separate because they fail for different reasons. The first is a person writing a form the paper does not hold, the second is a person walking past a form the paper does hold, and the third is the corpus losing a token on its way out of an automatic extractor.

The extractor counts are not a channel here. An extractor is written for one paper, and what it gets wrong is a fact about that paper and not a draw from a rate the next paper shares. Pooling 18 of them into one denominator would report a rate that nothing is sampling. They are in Section 5 per paper.

Channel	Failures	Trials	Bound
a form written that the paper does not hold	163	20261	0.00805
a token in the paper that no row holds	28	7440	0.00376
a token the corpus lost on the way out	4	9430	0.000424

A line has to pass all three. Taking them as independent, the joint bound is the product: **1.28e-08** per line

How much of that number has settled

Three significant figures is a format instead of a finding. The digit worth reporting has stopped moving, and the way to find it is to watch the bound as each paper joined the corpus.

Papers counted	Joint bound
1	2.468e-08
2	9.471e-09
3	5.946e-09
4	9.767e-10
5	6.013e-10
6	3.769e-10
7	2.426e-10
8	1.349e-10
9	9.373e-11
10	7.692e-11
11	6.587e-11
12	5.551e-11
13	4.947e-11
14	7.682e-09
15	7.073e-09
16	6.494e-09
17	1.284e-08

No digit has settled. The bound moves with every paper because every paper adds trials. Quoting 1.28e-08 to three figures would be reporting the format. The table shows the extraction still buying accuracy at the rate of about one order of magnitude every 59.9 papers, and the digit to report will settle when that stops.

Where 1e-26 lands

The target set for the extraction is 1e-26 per line over the whole extraction. It is not reached and it is not close, and the honest form of the answer is the distance.

Channel	Trials now	Trials at 993 papers	Bound then
a form written that the paper does not hold	20261	1183480	2.53e-06
a token in the paper that no row holds	7440	434583	6.9e-06
a token the corpus lost on the way out	9430	550822	5.45e-06

Reading all 993 papers of the archive, and finding nothing wrong in any of them, takes the joint bound from 1.28e-08 to about 9.53e-17. That is 10 orders of magnitude short of 1e-26.

Closing the rest by counting is not available. Each channel would have to reach about 1e+09 trials, which is roughly 1e+03 times the whole archive. There is no reading schedule that gets there, and a claim of 1e-26 from these three channels would be reporting a number nothing measured.

What a number that small would actually need is more channels that fail independently, not more trials in these three. Section 7 is one: a term recovered from the forms and scored against a border a linguist published, which fails for a reason none of the three share. Independent channels multiply, and that is how an exponent like this one is reached. Section 4 is where the independence is doubted, and it should be read before this number is quoted anywhere.

3.0.4 *4. What the number rests on*

The independence is the weak part and it is weak in three named places.

- **One person read every table.** The three channels catch different types of mistake but they do not catch a systematic misreading of one orthography, because the same reading produced the row and the expectation. This is the largest unmodeled term and no amount of trials touches it.
- **Direction one and direction two share a source.** Both ask questions of the same extracted text. A paper whose text is wrong in a way nobody has noticed fails both at once, and for that reason the papers with a known-damaged source are excluded from the count instead of given a worse bound.
- **Every channel runs through one implementation.** Two shared routines decide what counts as a language token, and all three channels ask them. A defect in either is common to all three at once. The references chapter documents two such defects.

What would move the number honestly is a second person reading a table that has already been read. That is the addition that fails for a reason none of the three share, and until it exists the first bullet is above every number in this chapter.

3.0.5 *5. Automatic extractors against their tables*

Paper	Rows	asked	for	Reproduced	Items	written	Invented	Wrong	language
[25]	443			443	1638		0	0	
[13]	52			50	323		1	0	
[24]	103			59	343		217	1	
[2]	682			261	1484		914	0	
[27]	285			256	337		80	0	
[18]	225			144	347		136	0	
[6]	369			219	1198		748	0	
[15]	1654			991	1625		574	362	
[9]	399			190	961		521	0	
[22]	1413			429	7154		2049	193	
[19]	653			369	4508		928	0	
[31]	193			101	435		323	0	
[33]	678			525	909		223	0	
[28]	288			152	533		337	0	
[23]	491			415	603		61	0	
[16]	295			228	423		124	0	
[26]	5753			1387	4039		2447	0	
[10]	468			299	777		117	0	

The extractors get a great deal wrong. The median reproduces 0.596 of what its table asks for, and the spread runs from one paper to the next with no common rate behind it, because each extractor was written to one paper's layout. These are a table in Section 3 instead of a term for that reason.

An extractor that does not reproduce a row is not by itself an impurity. The row is in the hand extraction either way, and the hand extraction is the test oracle. What the last two columns count is what the extractor added, the part that can reach the gold standard corpus without a person having written it.

3.0.6 *6. The word web*

The word web joins every form in the hand extractions to the forms it is related to, and groups them. The web has three types of edge, each measured off the extraction and none of them listed by hand.

- **concept**, two forms whose glosses share a content word. This is the edge that crosses an orthography, because the gloss is the only part of a form that two papers wrote the same way.
- **shape**, two forms of one group sharing a leading or trailing run of four characters. Salish morphology is heavily affixed and reduplicating. A shared run is usually a shared root or affix, and the shape edge measures that run without parsing it.
- **context**, two forms written in the same section of the same paper by the same speaker.

The web makes a language profile a concept expressed as a distribution instead of a bag of characters. The byte pair distribution cannot see that two orthographies wrote one word, and the concept edge is where that is recorded.

3.0.7 *7. The dialect border*

Lushootseed is not one dialect. The northern and southern varieties have known land and family borders, and Mellesmoen and Kye's stress paper labels every form it cites with which one it came from. The hand extraction copied that into the speaker column. The border is taken as a fact published by a linguist.

That makes it something a test of this algorithm almost never has: an answer that did not come from the algorithm. The border test loads the labels, sets them aside, and only compares at the end.

Comparing whole distributions does not work at this size. The method resolves at 6707 bytes and the two varieties hold 1469 and 2768. Neither the byte pair distribution nor the word web separates them, and a blind partition scores no better than the majority class. The check prints what that route would need: about 2.9 times the labeled Lushootseed now stored.

Asking for one term at a time does work, and it is the same move the sound work uses. Because both varieties are one language, nearly all of what a distribution over their runs measures is what they have in common, and at this size that shared mass swamps the difference. Flattening the pooled counts to max entropy takes it out: a run is weighed against where the pooled total alone would put it. A run carrying no border information then contributes nothing however common it is. Each run is its own test, and what a run needs is enough of itself instead of enough of the language.

The concept is retained fixed while this is asked, over the 58 concepts both varieties name, using the word web's gloss edge. Without that control a run can separate the two sets because the varieties differ or because different words were cited, and those are not the same finding.

Run	width	Runs	tested	Terms	found	Expected	by	chance	Random	borders	that	matched	it
2		75		2		0.20			1	of		200	
3		51		0		0.14			200	of		200	

4	23	0	0.06	200	of	200

Term	Deviate	Northern	Southern
ə́	-3.76	6	45
əs	3.18	6	0

The last column of the first table is the test that is allowed to fail. The border is put back on the same forms at random 200 times and the per-run test run again on each, and at width 2 only 1 of those random borders found as much as the published one. Swapping the two sides also negates every deviate exactly, but this estimator does that on any two sets whatever and it is evidence of nothing.

The term carrying the border is the stressed schwa, and it is southern: 45 of them to 6 in the north over the same concepts. The paper those labels came from is Mellesmoen and Kye's comparative analysis of stress in northern and southern Lushootseed. The algorithm was displayed the forms and never the labels, and what it returned is what the paper is about.

4

REPRODUCING WHAT THE LINGUISTS CLAIMED

4.1 NATER ON BELLA COOLA

This chapter takes a claim a linguist published about one of these languages and asks whether the corpus reproduces it.

Margaret Siwallace told the tale this is measured over. Hank Nater wrote it down and published it, and it was transcribed by hand for this work. The language is hers.

4.1.1 *Why this is a different type of test*

Almost every reading in this work is scored against the corpus it came out of. The dialect border of the chapter *How wrong it could be* is the exception that mattered max: a border a linguist published, withheld, and recovered. This is the second of that type and it is cheaper, because the answer was printed in a table forty years ago and nothing here can move it.

Nater's ICSNL 59 paper, *Voiceless Words in Bella Coola: Fact vs. Fiction* [28], disputes Mellesmoen (2021) on four points. Its Table 1 prints the Bella Coola phoneme inventory with a per centum distribution across the entire morpheme inventory, from Nater (1984:27). Thirty-four phonemes with a percentage each. Transcribed off the page they sum to 99.98. That sum is how the transcription is checked: a table with a row dropped still looks like a table, and one that sums to 100 does not have a row missing.

That table is a test oracle in the strict sense: a measurement somebody else made, over a sample this work has never seen, and no reading taken here can adjust it.

4.1.2 *What had to be thrown out first*

There are three Nuxalk corpora. Two of them cannot carry this measurement and neither says so in its name.

One is nearly a tenth English. The purity control is an English word list, words of four characters and up, drawn from the English reference corpus the screening already uses. *How Salish is Bella Coola* [26] scores 169 English tokens in 1934, at 0.087: *introduction*, *however*, *cognates*, *database*. The other two score 0.018 and 0.008.

The mark set is not the test used here, and using it would have been the obvious mistake. A mark set calls a token the language when it carries one of the characters the language is written with, and Ayajuthem and St'át'imcets are written in practical orthographies that use none of them.

Asked that way, Mary George's corpus reads 0.639 foreign and Carl Alexander's 0.602, when both are the language. An English word list has the opposite failure and it is the harmless one: it can only accuse a token of being English, and a Bella Coola word is not in an English dictionary.

One is Nater's own cluster charts. The gold standard corpus for the voiceless-words paper holds ck, cp, cq, kƛ'i: those are the entries of his Tables 2 to 5, lifted out of the paper by its automatic extractor. Counting phonemes over that file counts an obstruent cluster inventory as though it were speech, and pooling it with the narrative inflated every obstruent while leaving a five per cent short.

It is told from speech by measuring it. The tale runs 8.50 phonemes to a word with 0.954 of its words carrying a vowel; the chart runs under three phonemes with under a third of its entries vowelled. A chart has that shape and a language does not. The test is this measurement and not a list of excluded files, and the next chart to arrive is caught by it.

Margaret Siwallace's tale remains: 217 words, 644 phonemes read as a morpheme inventory, 31 characters no phoneme accounts for.

4.1.3 *The reproduction*

Counted as a morpheme inventory, the way Nater counted it, the tale sits at a total variation of 0.169 from his table.

Three of his rows read exactly zero in the corpus: the vocalic sonorants m̩, n̩ and l̩. The corpus does not write the syllabicity diacritic and defines m̩ as m. That is a fact about the extraction and not about Bella Coola, and it puts 3.58 per cent of the published table in cells the corpus can never fill while inflating the three plain sonorants by the same mass.

Folding his syllabic rows into their plain counterparts is a correction to the oracle. Doing it without saying so would make this worthless. Made openly, it predicts where the three plain sonorants should land, and they land there:

Phoneme	Nater, folded	The tale	Difference
m	4.99	5.59	+0.60
n	4.91	5.43	+0.52
l	4.82	4.50	-0.32

The total variation falls from 0.169 to **0.136**, and that fall of 0.033 measures what the three unwritten diacritics were costing.

Several rows agree closely without any correction at all. a, the commonest phoneme in the language, reads 15.06 against 15.84. i reads 7.92 against 8.12. ƛ' reads 0.93 against 0.95, xʷ 1.09 against 1.17, c' 1.86 against 1.78.

4.1.4 *Whether 0.136 is anything*

A total variation is a number until it has a floor under it and a wrong answer beside it.

The floor. The tale is split by alternating words, which samples it instead of cutting it in place, and the two halves are measured against each other. That is the resolution of this count at 644 phonemes: **0.066**. The reading sits at about twice its own floor. There is a real gap between this corpus and the published table and it is not all noise.

The wrong answer. The same count is taken over every other Salishan corpus stored and scored against Nater's Bella Coola table. Fourteen of them are large enough to ask.

Corpus	Phonemes	From Nater's table
Margaret Siwallace, Bella Coola	644	**0.136**
Delphine Derrickson Armstrong and Dave Michele	1001	0.239
Mary George	3295	0.299
George Lezard and others	3709	0.299
Martha Lamont and Annie Jack	550	0.315
Lottie Lindley	4138	0.318
Susie Sampson Peter and Martha LaMont	728	0.360
Carl Alexander	3106	0.392
wlwlmelst and Maurice Michell	1130	0.392
Charley Alexis Mayoos and William Celestin	373	0.397
Linda Redan	700	0.407
Kweltezetkwu and Bernice Garcia	1333	0.414
Bev Phillips	1530	0.444

The nearest language that is not Bella Coola sits at 0.239, and several of those corpora are five times the size of the tale. The gap from Bella Coola to the runner-up is 0.103, which is a floor and a half. A 217-word tale, extracted from a PDF automatically and checked by hand, lands nearer a phoneme table published in 1984 than fourteen other Salishan corpora do.

That is a positive control on the corpus and not on the method. It says that the extraction, the automatic extractor and the three checks preserved enough of Bella Coola for a frequency count to find its way home.

4.1.5 *The claim about nuclei*

Mellesmoen's Good Enough Nucleus Hypothesis, as Nater quotes it, holds that every word has at least one syllable and that anything equal or greater in sonority to a fricative may are in the nuclear position. Nater disputes the exclusion of stops, cites all-stop words and roots, and concludes with Newman (1947) that Bella Coola is largely non-syllabic.

Whether such words are in this corpus is a count. In Margaret Siwallace's tale, **17 distinct morphemes carry no sonorant and no vowel**: ck, cx, c'kt, kɬ, kʷc, kʷx̌', k'x, skʷ', sx, tx, tχ, tχʷ, t'xʷ, xɬ, xʷc among them. One of the 17 is built from stops alone: kʷkʷ'.

The cluster chart answers this question 103 times over and the answer is worth nothing, because every entry in it is an obstruent cluster by construction. Reciting the table a claim was argued from is not evidence for the claim. The narrative is the only line here that bears on it, and it rests on one short tale.

This counts occurrences and argues nothing about syllable theory, which nothing here reaches. What can be said is that voiceless words are in the running speech of the corpus, at 17 in a single tale. They are not an artifact of citation lists.

4.1.6 *What this cannot do*

The areal claim is out of reach. Nater's fourth point is that long obstruent clustering is shared with Oowekyala, at Rivers Inlet, and is therefore areal and not inherited. Oowekyala is Wakashan and there is no Oowekyala corpus here. The test has a shape: the same phoneme count over an Oowekyala corpus, against Bella Coola and against the Salishan corpora, with the question being whether Bella Coola sits nearer its neighbor than its family. Nothing here can run it until that corpus exists, and saying so is as far as this goes.

The residual is not explained. After the fold, s is 3.32 low, ʔ 2.01 high, u 2.36 high and kʷ 1.90 high. s is the commonest obstruent in the language and one 217-word tale is a thin sample of it.

Whether that gap is the sample, the genre, or a difference between Nater's morpheme inventory and any running narrative is not decided here, and more Bella Coola narrative would decide it.

A morpheme inventory built from marked boundaries is not the one Nater had. His was a lexicographer's list. This one holds what the corpus marks with a morpheme boundary, without duplicates. They are the same type of object and they are not the same object.

5

FROM THE PAGE TO A DISTRIBUTION OVER SOUNDS

This chapter follows one Salishan word from the mark on the page to a symbol a distribution can be taken over, and says which stage each measurement in this work belongs to. A measurement on this corpus is only interpretable once it is known what its symbols are for.

5.1 THREE STAGES, AND THEY ARE NOT INTERCHANGEABLE

A symbol in this corpus can mean three different things, and a distribution taken over each of them answers a different question.

1. A **character** is a codepoint in a file. A distribution over characters is a distribution over what a transcriber typed.
2. A **segment** is a base with everything that modifies it. A distribution over segments is a distribution over the units the orthography writes.
3. A **sound** is the referent of the segment, with the definitions of one sound collapsed. A distribution over sounds is the first one that is about the language.

Each stage discards something the stage before it carried, and each discard is a claim that has to be true. Skipping a stage does not leave the measurement neutral. It leaves it measuring the thing that stage was there to remove.

5.2 STAGE ZERO, THE PAGE

Nothing above starts until somebody has read the paper. A text extraction of a scanned typescript is not the paper: all four ICSNL 2 papers stored here [5, 11, 12, 17] carry none of their own orthography in their text layer, zero of the nine marks the corpus uses. They are listed as having no orthography in the text layer, so that no check grades a correct reading against a file the reading is right to differ from.

The trap runs the other way as well. *The Moon and the Birchbark Canoe* [7] is born digital and its text layer is faithful, but the embedded font draws U+026C, the l with a belt, as an l with a bar through the stem. A transcriber working from page renders sees ł and writes ł. The paper's text layer holds U+026C 112 times and U+0142 not once. The glyph on the page and the character the page is drawing are two different questions, and the hand extraction records the second with the first written down beside it.

5.3 STAGE ONE, CHARACTERS

The text space is the union of every character these orthographies write their languages with. It is built from the marked consonants and the schwa, the digit 7 of the practical orthographies, the stacked glottalization marks, the labialization modifier and Nater's raised space.

Two properties of it bear on every measurement.

It is a membership test and not an alphabet. Its job is to answer whether a token is in one of these languages. A per-paper set is written as the text space plus what that paper adds, never as that paper's own alphabet: a test built on one paper's characters reads every other paper as holding no language, and then reports nothing missing from them either.

It carries no phonetic content at all. ɬ and ƛ are both in it, as separate members, because both occur. Nothing in the text space says they are one sound.

Typographic ligatures are repaired before anything else, without a test, because ﬁ is fi and no decision is involved. Without that repair a case sensitive match counts five and ﬁve as different words.

5.4 STAGE TWO, SEGMENTS

ʷ is not a segment. It is labialization on the consonant before it. ʾ is not a segment either; it is glottalization on the base it sits over. Counting per character measures two halves of one segment as though they were independent.

The rule is to join every combining mark and every modifier character to the base in front of it. The hand extractions already do this when they write a form. Applied to Bev Phillips's *When Old One Created the Earth* [9], it turns 5552 characters over 35 distinct into 4839 segments over 50 distinct. The inventory grows under clustering instead of shrinking, because kʷ and k stop being one character counted twice. Twenty of the fifty carry a mark or a modifier:

ċ kʷ ḳ ḳʷ m̓ n̓ p̓ qʷ q̓ q̓ʷ w̓ xʷ x̣ x̣ʷ y̓ ẓ x̌ ə ʕʷ ʕ̓ʷ

ə is one of the fifty. It is the segment that carries the Lushootseed dialect border. At the character level it is two codepoints, and the per-run test finds it at width two for that reason.

5.5 STAGE THREE, SOUNDS

A correspondence table takes a definition and returns the sound it is for. The definitions that differ across the corpus, counted by segment over 23 hand extractions:

Sound	How the corpus writes it
lateral fricative	ɬ in seventeen papers, ƛ in four. LaFontaine and Janzen [18] write ƛ 329 times and ɬ never. Nater's voiceless paper writes ƛ 98 times and ɬ three.
glottal stop	ʔ in max, the digit 7 in the van Eijk orthography. Alexander and Davis [2] write 7 934 times and ʔ four. Matthewson and Redan [24] write 7 125 times and ʔ twice. A capital ʔ occurs in one paper.
uvular fricative	χ, x̌ and x, all three in six or more papers.
schwa	ə at U+0259 and ə at U+01DD, a turned e and a different codepoint.

Three of those four are named in the references chapter. The fourth comes out of counting, and so do the capital glottal stop and the aspirated pʰ and tʰ that only Nater writes. That is the argument for building the table by counting the corpus instead of from a description of it.

5.5.1 *Where the correspondences come from*

Nater states his own, in a paper this corpus holds [28]: `/ɬ/ = /ɬ/`, `/χ/ = /x̌/`, `/χʷ/ = /x̌ʷ/`, `/pʼ, tʼ, qʼ/ = /ṗ, ṫ, q̇/`, `/kʷʼ, qʷʼ/ = /k̇ʷ, q̇ʷ/` and `/ʼ/ = [ʔ]`. A linguist writing down what his own symbols equal is the strongest source available here. The van Eijk 7 is declared by its orthography. The rest follow the standard descriptions of Americanist transcription, they are assigned by hand, and any error in them is an error in every result taken downstream.

5.5.2 *Two collisions the table must avoid*

`kʷ` and `qʷ` must not go to one symbol. A labialized velar and a labialized uvular are different phonemes in these languages, and the collapse shows itself at once: `kʷɬ`, `qɬ` and `qʷɬ` arrive as one form, which merges three words.

The uvular fricative must not take the symbol `x`. A plain velar `x` passes through as that same symbol, and the mapped uvular and the unmapped velar then meet in one cell. A pass-through is not free of the namespace it passes into.

Every sound therefore takes a codepoint of its own from the private use area, assigned by the table and never typed. No sound can collide with a character the corpus writes. A table like this is exactly where a wrong assertion is invisible: nothing downstream can tell a real merger from a typo, and only reading its output for merged forms finds one.

5.5.3 *What the mapping does not claim*

Mapping `ɬ` and `ƛ` to one symbol says two definitions are one sound. That is a fact about transcription practice. Whether Nuxalk's lateral fricative and Lushootseed's are the same sound is a question for a comparativist, and nothing here asserts it. The mapping buys something narrow and it is enough: a difference measured afterward is not the difference between two typists.

5.6 WHAT A DISTRIBUTION READS AT EACH STAGE

Over what	What the number is about
bytes	the encoding, and whether a mark is precomposed or decomposed
characters	the transcriber, and which conference volume the paper appeared in
segments	the orthography, comparable inside one paper and across papers sharing a convention
sounds	the language, comparable across every paper in the corpus

The failure this ordering prevents has a definite shape. A model counting character bigrams reads two dialects that share a word as maximally far apart, because it is measuring the transcriber and not the speaker. The distance is real, it is stable, it will replicate, and it is about the wrong thing.

5.7 LIMITATIONS

The correspondence table is not part of the screening. The measurements of the other chapters take their distributions over bytes, and none of them is taken over sounds.

The clustering rule has a single implementation. Stage two is applied in one experiment and is not shared by the other measurements.

The word web is built on written forms. Nothing takes a character out of the text space and passes back what it is for at the point the web is built. Two dialects writing one word two ways are still two nodes. Stage three would close it.

6

THE RECORDINGS UNDER THE INSTRUMENT

6.1 WHAT THE PROSODY FIELD HOLDS

This chapter asks whether the prosody residual of a sound representation identifies the speaker, a question the orior workbook [30] leaves open, and reports what the representation holds instead.

Bev Phillips, of Lytton First Nation, spoke all three of these recordings, in nɬeʔkepmxcín, in the Lytton (x̌ʷa̓mcín) dialect. *When Old One Created the Earth* [9] is an original spíləx̣m of hers, written and told by her, and she is an author of that paper and of *You Will Be Sorry* [8] with Brent Hall. She chose the third story, *The Moon and the Birchbark Canoe* [7], and helped translate it, and Katherine Givens and Brent Hall transcribed and glossed it. The language is hers, and so are two of the three papers.

6.1.1 *The question*

The workbook states it and leaves it open: pitch against the speaker's own median is a per-speaker constant subtracted, the center of mass again. What remains in that field is the residual, and whether that residual identifies the speaker is the question.

The representation retains prosody apart from the segment code deliberately, because a field made shift invariant would put a stressed form and an unstressed one at one address. It holds four things per 10 ms frame: loudness, pitch against the speaker's own median, how that pitch is moving, and voicing. Each is thresholded at its own median over the recording. Each bit splits the frames in half by construction, and what a distance between two recordings reads is the joint arrangement of the four and never their levels.

6.1.2 *The recordings*

Recording	Seconds	Frames	Prosody, from flat
ICSNL 58, *The Moon and the Birchbark Canoe*	234.9	23489	0.3708
ICSNL 59, *You Will Be Sorry*	832.5	83249	0.3498
ICSNL 60, *When Old One Created the Earth*	1350.0	134997	0.4594

The 24 bit segment field reads 0.99 from flat on every one of them, and that figure is the frame count and not the speech: 16.8 million states against 134997 frames. The narrow widths are the ones to read.

These recordings are also at the length the vocalization work of the workbook says it needs. A one minute clip gives 1053 symbols for a whale and 281 for a bird, and measuring the arrangement of units needs tens of minutes. Two of these are 13.9 and 22.5 minutes.

6.1.3 *The answer, through the necessary condition*

Only one speaker is recorded, and there is no discrimination question to ask. The askable question is the condition underneath it: a residual that identifies a speaker has to hold still within one. Three recordings of Bev Phillips should then sit near the floor of the estimator.

A necessary condition cannot confirm identification, and its failure settles the matter. The thin sample is therefore enough.

Every recording is trimmed to 23489 frames, the shortest, because a null taken at one length bounds nothing about another. The floor is the recording split by alternating frames and measured against itself, which samples it instead of cutting it in place. A midpoint cut would put the first half of a narrative against the second and report the distance between two passages, the defect that inflates every split-half distance of the corpus chapter by 1.35 to 3.64 times.

		Apart
The Moon and the Birchbark Canoe	*When Old One Created the Earth*	0.1437
The Moon and the Birchbark Canoe	*You Will Be Sorry*	0.1807
You Will Be Sorry	*When Old One Created the Earth*	0.3112

The three floors are 0.0098, 0.0166 and 0.0077. The three distances run **9 to 19 floors**.

The residual does not hold still within one speaker. It does not identify her. One person, one language, one dialect, and her three recordings sit nine to nineteen times the resolution apart on the field built to hold what is left after her own pitch range is divided out.

6.1.4 *What that does not settle*

Three narratives in three ICSNL volumes vary together. The story, the session, the microphone and the encode all move with the recording and nothing here separates them. The measurement shows that the field does not hold still across them, and not which of those moved it.

The failure is also one directional in the useful way. Had the three sat at the floor, that would have been consistent with the residual identifying her and consistent with the field holding nothing at all, and a second speaker would have been needed to tell those apart. They do not sit at the floor, and no second speaker is needed to read that.

6.1.5 *Who spoke is read from the paper*

The speaker of each recording is taken from what its paper says and never from its filename. The filename of the ICSNL 58 recording names Givens and Hall, and the paper says the story was recounted in Nɬeʔkepmxcín by Bev Phillips, who also provided much of the translation, with Givens and Hall the linguists who transcribed and glossed it. A paper names its speaker in its prose, and a name put on somebody's words on no other evidence is a wrong attribution.

The design follows from that reading. Two speakers with one pair each give a comparison with a one in three ordering and no strength in it. One speaker across three recordings gives a necessary condition, and a necessary condition that fails settles the question.

6.2 THE SEGMENT FIELD IS NOT COMPARABLE BETWEEN RECORDINGS

6.2.1 *What the representation does*

The representation rotates the segment measurement before thresholding it, onto the right singular vectors of the centered measurement, and the measurement it is passed is one recording. Segment bit 3 of one recording and segment bit 3 of another are therefore different axes, and a distance between two recordings' segment profiles is partly a distance between two rotations.

The rotation exists because neighboring bands rise and fall together, and thresholding them as they come gives bits that restate each other. The basis it produces belongs to the recording.

6.2.2 *A comparison that does not isolate the rotation*

Comparing two halves of one recording, which share a rotation, against two whole recordings, which do not, gives ratios of 8.8 to 1.6 across the widths and looks like a confirmation.

It is not one. The prosody field is never rotated and gives 18.6 by the same measure. Two recordings differ for every other reason as well, and a check both arms pass is measuring something both arms have.

6.2.3 *What isolates it*

One recording, cut in half once, and the two halves compared twice. In the first arm the representation is computed on the whole recording and the frames are cut afterwards, and one rotation computed over both halves is used for both. In the second, each half of the samples is represented alone and carries a rotation computed over itself. The audio is identical and every other step is identical. The gap is the rotation.

Recording, at 6 bits	One basis	Two bases	The rotation
The Moon and the Birchbark Canoe	0.0995	0.3376	+0.2381
You Will Be Sorry	0.0769	0.2324	+0.1556
When Old One Created the Earth	0.0617	0.1430	+0.0813

At six bits the rotation multiplies the distance between two halves of one recording by 2.3 to 3.4 times. It costs +0.15 to +0.28 at eight, ten and twelve bits as well.

The measured distances between recordings' segment fields at six bits are 0.2309, 0.2346 and 0.2819, the same size as the rotation cost on its own. **A segment distance taken between two recordings is reading their bases.** The field is not comparable across recordings without a shared rotation, and no segment distance between recordings is reported here as a reading of speech. A comparison of two recordings' segment fields has to fit one rotation over both first.

6.2.4 *Why the prosody field escapes it*

It is not rotated: its four columns are each a named thing, and rotating them would mix pitch back into the field the segment code was made shift invariant to get away from. Loudness, pitch against the speaker's own median, the movement of that pitch and voicing mean the same thing in every recording.

The field that can carry the workbook's question is therefore the comparable one, and the answer it gives is negative.

6.3 THE WAVEFORM ITSELF, WITH NOTHING THROWN AWAY

6.3.1 *Why the representation cannot answer this*

The representation reduces a recording to 28 bits every 10 ms and discards the rest. It takes an envelope, smooths the harmonic comb off it, retains 24 coefficients, rotates them onto that recording's own singular vectors, and cuts every column at its median. Each of those is a bound, and a number taken through a bound describes the choice of bound as well as the data.

The decoded waveform has no such choice in it. All three recordings are written out at full integer width, in order, 215 MB of samples, and every reading below is taken off that. The bytes are checked against the corpus's signed audit first, because a measurement that moves for no reason is to be blamed on the bytes before the method.

6.3.2 *A linear slice reads the silence*

Reading the top w bits of each sample returns $H_2(2)$ equal to $H_2(1)$, and $H_2(16)$ at 5.04 bits, far under what a speech waveform should carry.

The recordings say why. *When Old One Created the Earth* peaks at 16954 of 32768 and its median absolute sample is **6**. Half of a twenty-two minute narrative sits inside thirteen levels of a sixty-five thousand level converter, because a told story is mostly the quiet between words. The top bits are sign extension: at two bits they take two values with the commonest holding 0.549 of the samples.

A linear slice of a waveform spends its levels where the signal is not. The workbook records the same defect four separate ways, with the correction.

Two seatings are therefore reported. Linear over the range each recording uses, whose H_2 curve is a fact about the amplitude distribution. And quantile, which puts an equal share of samples in every symbol, so that H_2 comes out at w by construction. The second is the point: the marginal is flattened to max entropy and every state is reachable. What is left to read afterward is arrangement and not loudness. It is the same subtraction the per-run test makes on two dialects.

6.3.3 *Where the arrangement sits*

The block shuffle retains everything shorter than its span and destroys everything longer. Sweeping the span locates the structure instead of only detecting it. The workbook builds this null on text; it needs recordings of tens of minutes to apply to a waveform. These are 3.9, 13.9 and 22.5.

Read on the quantile seating at eight bits, live against its own shuffle:

Span, seconds	*Moon*	*Sorry*	*Old One*	What that span is
0.00002	422.6	363.5	683.9	inside a glottal period
0.0002	48.0	46.9	71.8	inside a glottal period
0.0015	19.9	22.0	27.9	a pitch period
0.011	7.5	6.9	9.9	a segment
0.09	2.20	2.04	2.69	a phone
0.74	1.18	1.14	1.42	a word
5.9	2.83	1.05	1.07	a phrase

A ratio above one is the live recording carrying gaps more variable than its own shuffle, which is clustering. The workbook writes the same quantity as a ratio below one and the two are reciprocals. The *Moon* row at 5.9 seconds sits on 43 blocks and is scatter.

Three recordings, one curve, monotone across four orders of magnitude in span.

6.3.4 *There is no unit scale in it, and that absence is the result*

The obvious reading of that table is a knee somewhere near the word, and the text of one of these recordings is stored to check it against: 1736 words over 1350 seconds is 0.778 seconds a word, arrived at with no reference to the audio.

Swept finely between a phone and six seconds, there is no knee. The excess over the shuffle decays smoothly and fits a power law in the span:

$$\text{excess} \;\propto\; \text{span}^{-0.752}, \qquad r^2 = 0.988$$

over a 64-fold range of spans, with every one of eleven points landing between 0.88 and 1.19 of the fit. Nothing happens at 0.778 seconds. Nothing happens anywhere.

The waveform's arrangement has no characteristic scale. The absence is itself the measurement, and it is the case the workbook names: data carrying structure at every scale has no characteristic scale for a window, a band or a ceiling to sit at. What holds still for such data is how the quantity changes with scale, and here that is an exponent measured to $r^2 = 0.988$.

6.3.5 *What that settles about reading speech back*

Detecting that a recording is speech is available and is not the interesting half. The departure from the null runs 363 to 684 at sample scale and is still 1.07 at six seconds. Nothing this large needs a careful instrument. The majority of it is the amplitude envelope, which is to say the silence between words, and a measure reading that is reading that a person paused.

Transcription is not available from this reading, and the reason is structural. Writing down what was said requires knowing where one unit ends and the next begins. The method finds a boundary in text by a symbol whose gaps are far tighter than geometric [29], and the whole of that construction rests on there being a scale where the regularity sits. This waveform has no

such scale. A power law with no knee supplies no unit boundary at any span. There is nothing to segment on and nothing to write down.

The same obstacle is from the other side: the null for animal vocalization has to retain each call intact while permuting the order of calls. That null requires fixing the unit, and the unit is the thing being asked for. The block shuffle is the closest available substitute and it returns no unit.

Dialect cannot be asked at all. The audio stored here is one speaker, one language, one dialect: Bev Phillips, nɬeʔkepmxcín, Lytton. There is no second dialect to discern from the first, and a summed reference over one dialect is that dialect. The dialect question has an answer in this work and it is on the text side, where the published labels of Mellesmoen and Kye [25] supply a border the algorithm never sees. Asking it of sound needs recordings carrying a border somebody else labeled, and the corpus holds none.

6.4 THE JOIN BETWEEN THE WEB AND THE SOUND

6.4.1 *The missing stage*

The chapter *From the page to a distribution over sounds* states the three stages and says which one is absent: the word web is built on written forms, and nothing takes a character out of the text space and passes back what it is for. Two dialects writing one word two ways are still two nodes.

The section above looks for the unit in the waveform alone and finds a power law with no knee. That is the middle of the three ways a partition is fixed, an estimate, and an estimate reaches nothing past what the sample already carries. The third way adds information the sample does not hold, and it is the only one that does.

6.4.2 *The supervision is printed on the page*

Hall and Phillips [9] write that there is no standardized punctuation convention for nɬeʔkepmxcín, and instead of periods they separate each sentence of the transcription with a timestamp into the recording. The hand extraction carries those timestamps in its gloss column.

That is **126 sentences of Bev Phillips's own spíləx̣m, each carrying the interval of audio it was spoken in and the exact characters she said**, and no algorithm produced any of it. It sits in the gloss column of a table checked against its paper in both directions.

It is checked before it is used: the timestamps run forward, the last is 1336 s against a 1350 s recording, and the 126 sentences cover all of it. A sentence runs 2 to 40 seconds, median 10.

6.4.3 *A segment is a grapheme cluster and not a character*

The segments are formed by the clustering rule of the chapter *From the page to a distribution over sounds*: every mark and modifier joined to the base in front of it. On this paper that gives **4839 segments over 50 distinct**. ə́ is one of the fifty, at 45 instances, the segment that carries the Lushootseed border.

6.4.4 *Placing a segment inside its sentence, and why the crude answer is allowed*

A timestamp bounds a sentence and not a segment. The segments inside one are placed evenly, which is wrong in detail, since a vowel is longer than a stop and nothing here knows that.

The comparison it is checked against allows it. The null permutes which character is attached to which interval and leaves the rest unchanged. Every interval retains its position, its length and its audio, and the even spacing sits in both arms exactly. An error in the placement costs both arms the same and cannot manufacture a result. It also bounds the finding: what reads as holding together is holding together at the resolution even spacing allows.

6.4.5 *What the pooled sound gives*

Each interval becomes the loudness heard in 64 log spaced bands, normalized so that a loud instance and a quiet one of one segment are one shape. A segment type's sound is the mean over its instances, and that is their sum. Thirty-four of the fifty types carry at least 24 instances.

a type's instances sit from their own mean	0.2766
two types' means sit apart	0.0591
between over within	0.214
the same characters shuffled over the same intervals, median	0.191
shuffles reaching the real characters	**9 of 200**

Two things have to be read off that together, and reading either alone gets it wrong.

The alignment is real. Nine of two hundred shuffles reach what the characters a person wrote down give. The characters and the audio are not independent, and the only thing joining them is a timestamp printed in a paper.

The characters do not separate. Between over within is 0.214, which is to say a segment's own instances scatter about five times further than one segment's mean sits from another's. A representation that recovered a character from its sound would have that the other way up. This one does not, at this resolution.

The join exists and carries something, and it is nowhere near reading a character off the sound. The stage between written form and sound is measured, and the number on it is small.

6.4.6 *What it needs next, which the measurement names*

Even placement is the binding limit and it is fixable without giving up the supervision. The characters would still come from the hand extraction and only the boundaries would come from the audio, and the null would still permute labels over whatever boundaries were found. The control does not weaken. A sentence of ten seconds holding thirty-eight segments gives each one 0.26 s where a stop is nearer 0.05. Max intervals straddle a boundary or land in silence. That is where the scatter is.

What must not be done is to let the audio choose the characters. The timestamps and the transcription are everything that makes this supervision and not an estimate, and an alignment tuned until the characters separate would be the algorithm scoring itself against a split it proposed, a test that passes whatever the data.

7

MATERIALS, PROCEDURE AND AVAILABILITY

7.1 SOURCE TEXTS

The papers are the published proceedings of the International Conference on Salish and Neighbouring Languages, distributed as PDF files by the University of British Columbia Working Papers in Linguistics. The archive holds 993 papers; the automatic extractors need eleven of them, and the screening reads every paper with no extractor.

Two conversion choices have a wrong answer that looks like a right one.

The encoding. A PDF-to-text converter that writes Latin-1 has neither ʔ, ə nor ɬ. A paper converted that way still opens and still looks like a paper, with the language taken out of it. Every paper here is converted to UTF-8.

The reading order. Converted in layout order, a running header prints above the title it sits under and a two-column table interleaves. The papers are read in the order the PDF stores the page, and the automatic extractors are written against that order.

A converted paper whose fonts renumber their glyph codes and declare no map back to Unicode is marked as not representing the page, with the fonts named. Its text is retained and nothing downstream reads it as the paper. Six of the 146 PDFs stored are in that class, and the references chapter names them.

7.2 WHEN THE TEXT IS NOT THE PAGE

Two of the eleven papers, Lyon [22] and Lindley and Lyon [19], are set in a font of that class: their text layer holds the font's alphabet and not what the page prints. For those two each page is rendered as an image at three times its size, which puts a 12 pt body around 50 pixels tall, where a stacked diacritic stops being a guess. A page text is drafted from the font's codes, line for line with the extraction, and graded against the hand extraction. Two questions cannot be settled without looking at the page: which `w` is a labialized consonant, and which inserted space is a word boundary.

The drafted page text replaces the font repair for those papers. Applying the mapping to text it has already been applied to destroys the text.

7.3 PROCEDURE

1. Convert each paper to text, as above.

2. For the two papers whose text is not the page, render the pages and draft a page text.

3. Run each paper's automatic extractor, which writes that paper's gold standard lines, its other text, and the lines it could not sort.

4. Check each hand extraction against its paper in both directions. A form written down that the paper does not hold is a typing slip. A word in the paper that no row holds is a row somebody skipped. A person transcribing a thirty-seven page paper into a table produces both. For the two drafted papers, the first direction measures the draft and not the table.

5. Check each automatic extractor against the hand extraction: whether it found each form, whether it assigns the same type, and whether it assigns the same dialect. Type decides the corpus. A rejected tableau candidate filed as a citation form puts a form the paper's own analysis rejects into the gold standard corpus.

6. Check coverage: every language token in each paper against the corpus built from it, both sides through the same repairs. For the two drafted papers a coverage of zero missing means the extractor got everything the font's encoding contained, which is weaker.

7. Calibrate the screening against the gold standard corpora and the English spans the hand extractions mark, then screen every paper without an extractor against both profiles.

8. Read which language each paper is about out of its own title and abstract, ask whether the distributions agree, and admit candidates to each corpus in batches while it stays on its own growth curve.

Two further tests support the repairs. A candidate character mapping is tested by counting how many damaged tokens become forms attested in a clean paper; the reading is the change between the two rates. The screened output is matched case sensitively against the gold standard corpus, and a form that matches only once case is folded is the same word written wrong; the count of those is the formatting damage.

7.4 SHARED DEFINITIONS

One definition, the text space, fixes which characters these orthographies are written with, and every component reads it. A per-paper set is the text space plus what that paper adds. The repairs are each one function: putting back words split internally by an extraction, closing the space a PDF leaves after a stacked diacritic (with the mark set each paper's spacing allows), reading a paper's glyph names back as the characters they name, and rejoining words broken across lines.

7.5 THE SOUND REPRESENTATION

Each recording is reduced to one row per 10 ms frame: a 24 bit segment field and a 4 bit prosody field. The segment field is the loudness heard in log spaced bands, made shift invariant and rotated onto the recording's own singular vectors; the prosody field is loudness, pitch against the speaker's median, pitch movement and voicing. The delta from a flat distribution is read at 6, 8,

10, 12, 14 and 24 bits. The narrow widths are the ones to read: a 24 bit field has 16.8 million states and the longest recording has 135 thousand frames.

7.6 DATA AND CODE AVAILABILITY

The source papers and recordings are published by UBCWPL and are available from its archive. The hand extractions, the automatic extractors, the checks, the screening and the sound representation are in the orior repository, `https://github.com/dstroy0/orior`. Every number in this research paper is computed by that code from those inputs.

8

SOURCES

8.1 SPEAKERS

These languages belong to the people who speak them, and none of this work exists without them. The corpus everything else here is measured against is their words, written down.

The chapter listing the gold standard corpus opens every entry with the speaker and carries the conditions they set. A speaker's name is stated there once and nowhere else.

The majority of the papers name their speakers. The rest cite a published dictionary and never say who spoke, and their entries say so. A linguist wrote the paper and a person transcribed the paper into a table, and neither of those is whose language it is.

8.2 PAPERS WITH AN AUTOMATIC EXTRACTOR

Each of the papers below has an automatic extractor written for it and a hand extraction beside it. Every token of the language in each is accounted for in the corpus built from it, by the coverage check. For the two Lyon papers that statement is about the extracted text, the font's own encoding, and not what the page prints; the section below on those two says what that means.

Paper	Language
Three Glossed Nɬeʔkepmxcín Narratives by Kʷəɬtəzétkʷu (Bernice Garcia) [6]	nɬeʔkepmxcín
ɬ cutés us ɬ qəɬmín ɬ tmíxʷ (When Old One Created the Earth) [9]	nɬeʔkepmxcín
Four Stories by wlwlmelst [18]	nɬeʔkepmxcín
Cw7aoz káti7 láti7 ku naxwít [24]	St'át'imcets
I Tsícwas sQwa7yán'ak Áku7 Graveyard Valley [2]	St'át'imcets
Mary George Personal Narratives [15]	St'át'imcets
A Bella Coola tale: The Frog Children [27]	Nuxalk
Three Okanagan stories about priests [22]	Nsyilxcən
12 more Upper Nicola Okanagan narratives [19]	Nsyilxcən
Poking Fun in Lushootseed [13]	Lushootseed
A Comparative Analysis of Stress in Northern and Southern Lushootseed [25]	Lushootseed
BC Indigenous people's Chinuk pipa script: History, analysis, and texts [31]	nɬeʔkepmxcín, Secwepemctsín
Lexical Suffixes and Connectives in Proto-Central Salish and Beyond [33]	eighteen languages
Voiceless Words in Bella Coola: Fact vs. Fiction [28]	Nuxalk
Nsyilxcn Inchoatives and their Distributions Across Root Types [23]	Nsyilxcən
The truncated reduplication in Twana [16]	Twana
How Salish is Bella Coola? [26]	Nuxalk
Ctrl-Alt-Delete: The Control Directive and Associated /t/ Deletion in nɬeʔkepmxcín [10]	nɬeʔkepmxcín

Each automatic extractor writes the marked record, the gold standard lines holding only target-language speech, a list of what it could not classify, and for the two Lyon papers the word forms recovered from the interlinear. A record is named speaker first, because the speaker is who the corpus is of.

The papers differ in what makes them hard to extract, and the differences are the evidence for how each extractor is built.

8.2.1 *Surface and underlying forms*

Hall, Luntzlara, Mellesmoen and Reid print every form twice, and only one of the two was said [10]. They argue about what the nɬeʔkepmxcín transitive suffixes are underlyingly. Each example carries a surface form in square brackets and the underlying form they propose for it in slashes: `[wéc̓entis]` beside `/wéc̓e-n-t-ey-es/`. The hand extraction is 494 rows and verifies in both directions with nothing left over.

That layout makes the delimiter the evidence, and the extractor reads only the delimiter. A bracketed run is a word and reaches the gold standard corpus. A slashed run is the authors' analysis, holds morpheme boundaries and a null-morpheme sign, and does not. A star marks a form the analysis predicts and the language does not have, written `*[kícetxʷ]` in one place and `[*kícexʷ]` in another, and the extractor tests for both shapes. The delimiters stay on the form in the record, because the record holds what the page printed, and come off on the way to the gold standard corpus, where the entry is the word.

The extractor does not find two types of form, by design. Examples (24) and (29) to (31) are ordered-rule derivations and every line between the underlying form and the surface form is a stage: `kícntene`, `kícntne`, `kícnne`, `kícne`. Those carry no delimiter, because the paper sets them in a column. The prose also prints some surface forms bare, `kicnəxʷ` and `wíktxʷ` among them. The hand extraction has all of them and the extractor leaves them unclassified, which for a form nobody said is the safe direction to be wrong in.

Section 3.1.1 argues from two other languages, and the speaker column carries them: ʔayʔaǰuθəm has `-θi` and St'át'imcets has `-ci`, with Newman's proto-Salish `*c` and `*ci` cited beside them. The extractor puts zero forms under the wrong language. One string is genuinely ambiguous: `-ci` is both the St'át'imcets suffix the paper cites and the nɬeʔkepmxcín suffix the paper proposes, and nothing in the token separates them. `-ci` enters as nɬeʔkepmxcín, which the paper argues it is.

The extractor removes matched spans by position and not by name. Page 13 carries `/ʔ/`, which yields the one-character form `ʔ`, and removing that by name would also remove it from `{n, ṅ, h, ʔ}]morpheme-t-___`. One token does not come back through the coverage check: `ʔ}]morpheme-t-___` holds a brace, the record delimits its spans with braces, and the check splits it on the way back out. The token is in the paper and in the record, and the paper is at 99.7 percent coverage for it.

8.2.2 *An etymological lexicon*

Nater's *How Salish is Bella Coola?* is the largest table in the set [26]. Nater asks how Salish Bella Coola is, and his answer is his whole lexicon sorted by where each word came from: 209 proto-Salish, 94 Coastal, 63 Interior, 214 non-Salish, 34 areal, and 661 of unknown origin. That is 1407 numbered entries over 52 pages, more of one language than the rest of this set holds together.

The fourth column of its two tables is whichever language Nater is comparing against, and the extractor finds where that column starts by the token that opens it: one of the sixteen abbreviations section 1.1 defines, a language the body defines, or a reconstruction opening with an asterisk. Haisla, Heiltsuk, Oowekyala and Kwak̓wala are North Wakashan. Proto-Athabascan, Eyak, Carrier and Tahltan are Na-Dene. Nootka, Quileute, Chinook and Yurok are four families past that.

Squamish, Sechelt, Shuswap, Lillooet and Halkomelem are Salish and still not this language. The speaker column carries all of it, and only Bella Coola reaches the gold standard corpus.

The paper prints that column's own language at the head of every table page, as `Line Gloss BC CS Cognates`, and the extractor reads it from there. A bare `*ƛ̓əp` under that head is Coast Salish and holds nothing in itself that an explicit tag would have supplied. Reading the head places 326 reconstructions under the name of the group they reconstruct.

Section 4 prints each of its entries twice. `AKW'A` and `ʔakʷa` are one word, and section 2.1 gives the mapping: `ʷ` is written `w`, `c` is `ts`, `ƛ̓` is `tl̓`, `ɬ` is `lh`, `x` is `c`, `χ` is `x`, and `ʔ` is `7`. Both are Bella Coola and the table records both. Only the phonemic column reaches the gold standard corpus, because two orthographies in one stream would put the byte pairs of a transliteration into the measurement of a language.

One paper can write one sound two ways. The tables set every ejective as `̓`, U+201F, 1763 times. Section 3's prose sets it as `'`, U+2019, 51 times. The prose's `t'nχʷ` and the appendix's `t̓nχʷ` are one word in two encodings. The schwa does the same, `ə` U+01DD 183 times against `ə` U+0259 52 times, and NFC unifies neither pair. A mark set built from the tables alone reads section 3 as English.

Its practical orthography also writes the glottal stop as the digit `7`, which the shared mark set holds. A citation like `Sq67:137` or `Ku02:86-87` therefore carries a character of the language, and the second direction of the check asks about it. A digit that is a character in one orthography cannot also be punctuation in a check that covers that orthography.

The table records each cognate as its own row, and the extractor writes the cognate column as the single string the paper printed: `*ƛ̓əp „deep (water)‟` is one item against two rows. The extractor's large counts of forms not found and forms added are that one structural difference counted from two sides.

8.2.3 *Twana*

Kim's paper on Twana reduplication [16] is the only Twana source here. Its hand extraction is 358 rows. Every Twana form in it is Drachman's, out of a 1969 dissertation the paper calls the only reliable reference in existence for Twana CVC reduplication. What is on these pages is close to everything written down.

Kim also carries the trap this set makes easiest to fall into. Example (1) sits where the data usually starts and it is Tillamook, in Edel's 1939 transcription with its own vowel characters; (19) and (20) are Thompson and Lillooet; footnote 9 mixes Puget Sound Salish, Moses-Columbian and a row of English phonetics. The speaker column carries all of it and the footnote is left unclassified, because the language changes inside one sentence there.

The lateral fricative has a third character. Papers differ between `ɬ` and `ɮ`, and the text space carries both. Kim writes it `ɮ`, U+026B, sixty-two times, and a mark set holding the other two reads every one of those tokens as English. The same paper writes the glottal stop two ways, `ʔ` where it is phonemic and `ˀ` where a rule derived it, which its footnote 11 states outright.

The extraction doubles a combining mark on four phonetic forms. Those four are corrected at the source, and the table records the form the page prints; transcribing the doubled form would put a corrupt string in the table.

8.2.4 *Elicited roots*

Lyon's survey of how 48 Nsyilxcn roots inchoativize [23] is elicited from ɬk̓mxnalqs Delphine Derrickson-Armstrong and c̓əsqʕákna? Dave Michele of stq̓a?tkʷɬniẇt. Its hand extraction is 537 rows and verifies in both directions. Max cells of its two tables are starred, a form the linguist built and the speakers rejected, and those are excluded from the gold standard corpus: a smaller corpus is better than one seeded with words the language does not have. The extractor files the Spokane, Secwepemctsín, Lillooet, nxa?amxčín, Thompson and Coeur d'Alene cognates under their own languages.

The paper is at 99.5 percent coverage, and neither of the two missing tokens is a word. Closing the space after a glottalization mark welds √puɬ and púɬ on page 9 onto the `qt:{` that opens the dictionary quotation after them. The welded tokens hold a brace, the record delimits its spans with braces, and the coverage check splits them on the way back out. They exist in neither the page nor any corpus.

8.2.5 *Words with no vowel*

Nater's *Voiceless Words in Bella Coola* [28] argues that Bella Coola has words with no vowel in them, and its data is 127 numbered entries that are each a run of two to four consonants. Six more entries and two comparison tables are Heiltsuk, Oowekyala, Kwak̓wala and Haisla, which are North Wakashan and not Salish at all, and the speaker column carries the distinction. Its hand extraction is 671 rows and verifies in both directions with nothing left over.

8.2.6 *A comparative reconstruction*

Wolfe's paper [33] is a comparative reconstruction. Instead of one speaker telling one story it cites lexical suffixes from published dictionaries of eighteen languages, arranged in columns under a two-character abbreviation. Its hand extraction is 772 rows and verifies in both directions with nothing left over. Its wrong-language count carries the max weight of any paper here, because it holds more than two languages, and it reads zero: the speaker column carries the language on every row and the extractor keys its output to the same column.

8.3 HAND EXTRACTION

The automatic extractor is not the source of the corpus. Each paper is transcribed by a person into a table that says, for every form in the paper, what it is, and the extractor is graded against that table. The table pairs each form with its meaning; it is a glossed word list, and it is the research's own data.

Every table is checked against its paper in both directions: no form written that the paper does not hold, and no language token in the paper that no row covers. Of the 25 tables, 14 verify with nothing left over in either direction. Of the rest:

- two are checked against a drafted page text, because their extracted text is not the page, and what their disagreements measure is the damage in that text (the next section);

- one, Robertson [31], is damaged four ways in its extraction, and its disagreements are the same finding counted from both sides;
- five carry none of their own orthography in their text layer, the scanned 1975 typescript and the four ICSNL 2 papers, and are not counted;
- three carry the failures the chapter *How wrong it could be* counts: Givens and Hall [7] with 80 forms the paper does not hold, Nater's lexicon [26] with 81, and Alexander and Davis [2] with 2.

How each automatic extractor fares against its table is reported per paper in the chapter *How wrong it could be*, the only form in which those counts mean anything.

One hole in the second direction is not a hole in a table. Nater's argument is that Bella Coola has words with no vowel in them. Entries like `kp` 'each, all, every', `tp` 'spotted', `ps`, `px` and `sx` are plain ASCII with nothing in them that any mark set can catch. The language-token test needs a marked character, and about thirty of the 133 entries are invisible to the second direction. The first direction still checks every one of them, because it looks for what the table wrote down. The apostrophe cannot be the way in: Nater writes every ejective with one, but a gloss ends in a closing quote and the two characters are the same, and every English gloss then reads as a word of the language. That trade is a few hundred false holes for about thirty real ones, and the apostrophe stays out.

Verification separates the author from the speaker. In Hilbert's paper [13] the coverage check is 100 percent, and the hand extraction still corrects two things. Vi Hilbert wrote the paper; the twenty-one examples were said by her aunt **Susie Sampson Peter** of the Upper Skagit and by **Martha LaMont**, recorded by Leon Metcalf between 1950 and 1958 and by Thom Hess in 1963. And example 2's English translation is a single line, "High class, high class was Raven." The extraction welds the next eleven lines of Hilbert's commentary onto it. The typescript sets examples in an indented column and the essay at full width; the extraction drops the indent and the width recovers it.

Six of the PDFs break words in half. They leave a space after a stacked diacritic: `ḵweswapáẇ` arrives as `ḵ` and `weswapáẇ`. The counts are Hall and Phillips 996, Garcia 943, LaFontaine and Janzen 478, Mellesmoen and Kye 298, Matthewson and Redan 169, Mary George 159. The coverage check cannot see it, because it puts both sides through the same repair and a word broken on both sides matches itself. A repair closes it; `ʷ` is excluded, because it is a spacing character and a space after one is a real boundary.

The stress accents are excluded from that repair. A word can end in a stressed vowel, and `ntes neʔé e sqyéytn` in LaFontaine and Janzen would close to `neʔée`, which the language does not have. Mellesmoen and Kye passes its own wider set, because that PDF prints two spaces at a real boundary and a lone space after an acute there is always the inserted one.

Matthewson and Redan carries a second form of the same damage that no rule can close safely. The PDF also breaks *before* a marked character: `các̓lep` arrives as `các ̓lep`. The same shape is a real word boundary at `lta q̓íl̓qa`, and nothing in the characters separates the two. The extractor leaves both alone; the hand extraction has the true forms.

The apostrophe is a character. `'` is glottalization in Nuxalk and in Lyon's Okanagan, and stripping it as punctuation turns the enclitic `_c'` into `_c`.

8.4 PAPERS WHOSE EXTRACTED TEXT IS NOT THE PAGE

Six of the 146 PDFs stored carry a font that renumbers its glyph codes with an `/Encoding /Differences` array and declares no `/ToUnicode` map. An extractor is then passed code numbers with nothing to turn them into, reads them in a default encoding, and what can land in the text is the font's private alphabet.

Each of the six is opened against its own rendered pages, and they come out four different ways.

Paper	Fonts declaring no map back to Unicode	What the page shows
Lyon [22]	NimbusRomNo9L Regu, Medi, ReguItal	damaged, and transcribed from the pages in full
Lindley and Lyon [19]	NimbusRomNo9L Regu, Medi, ReguItal	damaged the same way, and transcribed from the pages in full
Lyon [21]	NimbusRomNo9L Regu, Medi, MediItal, ReguItal	damaged, by a different table
Robertson [31]	Symbol and five TrueType subsets	damaged four ways, one of them decodable exactly
Abraham [1]	NimbusRomNo9L Regu, Medi	clean
Lonsdale and Matsushita [20]	Courier, Times-Roman, Times-Italic, CMSY10	clean

A missing `/ToUnicode` is not the fault by itself. 141 of the 146 hold a font without one and nearly all of them extract correctly, because a standard encoding already says what the codes mean and every extractor has that table. It is the renumbering that does the damage, and 6 of those 141 renumber.

Renumbering does not always damage the text. Two of the six extract correctly, and both are checked line for line against a rendered page. Abraham's paper is St'át'imc in the van Eijk orthography, which is ASCII apart from the accented vowels, and its renumbered codes never reach a character they would damage: page 1's first line reads `Ats'xenlhkán ta sásqets áku7 Nséq'a (Charlie Mack's), lti Líl'wata Tsel'álh c.walh,` on the page and in the text. Lonsdale and Matsushita print their Lushootseed in the ASCII transliteration their own parser reads, and there is nothing on the page for a font to lose. Page 5 sets `LEFT-WALL ?u+ da?a +d ?ElgWE? ?E kWi s+ gWistalb ti?E? SukWE? .` and the extraction gives that string exactly.

Page 23 of Lindley and Lyon prints

`cítxʷsəlx uɬ ṫi nyʕip cḵaʔítət`

and the extracted text holds

`cítxws@lx uì 'ti ny 'Qip c 'kaPít@t`

Every schwa is `@`, every lateral fricative `ì`, every pharyngeal `Q`, every glottal stop `P`; the wedge is in front of its character as `ˇx`, the ejective mark is in front of its character instead of over it, and the word `cḵaʔítət` is split. Two independent PDF libraries lose the same things. This is then a property of the file and not of one library.

One part of the loss cannot be inverted. Labialization is written with a raised w and the page also has a plain w. Both arrive as `w`. Page `kʷukʷ` and page `wist` are the same string in the text, and nothing in the file separates them.

The consequence for method is the reason this section exists. A hand extraction taken from one of these texts records the font's encoding and not the paper, and it then verifies clean against that same text in both directions, because both sides of the check are reading the one damaged artifact. The verification of these papers is therefore done against the page.

The pages are rendered as images and the transcription is done from the image. A page text is drafted from the font's codes in the shared orthography, line for line with the extraction, and a page of one is then a page of the other. The draft is a starting point a person corrects against the rendered page, never a source. There are seven things it cannot decide:

- **Labialization**, for the reason above. The drafter writes ʷ after the consonants that take one, which is right for kʷ qʷ xʷ x̌ʷ and wrong wherever a real w follows a consonant. The rule errs in both directions, and page 315 has one of each. púlxwi gets a ʷ it should not have, and section 3.2 settles that by parsing the word √púlx-wi. lkʷu·····t loses the one it should have, because the inserted space splits it into lk and wu·····t before the rule can see the consonant, and section 3.2 parses that one √lkʷ=ut. The rule runs over the whole line and reaches the English as well: stanza 117's word gloss he.fell.off.backwards comes out he.fell.off.backʷards.

- **Word boundaries.** The PDF inserts a space in front of a character carrying a mark. s ˈplá ˈks@lx is one word and iP ˈkl is two, and both are a space in front of a marked character. Page 25 settles the first as sp̓lák̓səlx, page 24 the second as i? k̓l.

- **The space that was not inserted.** Where the PDF leaves the mark against its character there is nothing to key on, and a medial ˈ is also the apostrophe of Lyon's. nkwˈritkw, xwˈtilx and waˈy retain the mark located in front of its character; the pages read them nkʷr̓itkʷ, xʷt̓ilx and way̓.

- **A P-initial word with no other mark on it.** Pitx, Pums, Pistkm and Paksyilx come through as English, because a capital P opening an otherwise unmarked token is the shape of Papers and Penticton.

- **The dropped space.** At a change of font and at some line ends the PDF loses the space entirely. Okanagan yaxʷt, uɬ ny, məɬ ixí? and axá? ikɬcítxʷ arrive welded into one token.

- **A run of the length mark before a closing quote.** The raised dot arrives as ;, and a rule turns it back wherever a character or a spacing mark follows it, because a lone ; is also Lyon's semicolon. A run of eight is never punctuation, and sta?x̌íləm "whoa·····" is the only place where a run is followed by a closing quote instead. That pattern occurs twice, both in the priests paper.

- **A one-character gloss label that is also a glyph code.** A gloss token is recognized by a run of two capitals, because one capital on its own is also iP and Philosophical. Stanza 108's gloss line labels the question marker Q, one capital, and the token comes out ʔ. Nothing inside the token separates the two readings. That label occurs twice, both in the priests paper.

Every disagreement the check reports on Lindley and Lyon falls in these classes, and the last two occur in the priests paper alone. That count is the measurement the draft exists to produce.

The extraction is better evidence than the rendered page for one thing. The transposition moves a mark off its character and preserves it, and a token's mark count survives the channel exactly. At the scale these pages render, a run like m̓y̓m̓y̓á cannot be told from m̓ym̓á by eye, and reading the page alone undercounts. Stanza 112 arrives as s- m̓ y̓- m̓ y̓-á y̓-s, which is five marks, and the word is s-m̓y̓-m̓y̓-áy̓-s. The count is taken from the text and the boundaries from the page.

8.4.1 *A scanned typescript is a worse case than any font*

Hilbert and Hess [14] carries no font problem at all. It is a 1975 typescript, scanned, and its text is OCR of the scan. Page 1 prints

kʷa? ləcuxəčadadid dxʷ?al ti?ə? s?acuss (h)əlgʷə? dəxʷxəbil ?ə ti?ə?.

and the text holds

kWa? lacuXacadadid dxW?al ti?a? s?acuss (h)algWa? daxwxabil ?a ti?a?

Every schwa is a, every raised ʷ is W, the wedge over x is gone, and sudᶻigʷicuts comes through as sudZigWiOltS.

The difference from the Lyon papers matters more than the size of it. There the damage is a font's renumbering, which is deterministic: one code always is for one glyph, which lets a table invert it. Here the damage is OCR guessing at shapes on a scan, and sudᶻigʷicuts to sudZigWiOltS is not a substitution anything can run backwards. The hand extraction is not a check on the extraction for this paper; it is the only correct record of it.

That also puts a floor under the whole method. A paper this old cannot be verified in both directions, because one direction has no sound source to verify against. What the check reports on it is the distance between a page and its OCR, which is worth having and is not the same measurement it makes elsewhere.

The check strips ? as punctuation, and this orthography and the 1983 typescript's both write the glottal stop as ?. ?ə bares to ə, and a word with a glottal stop cannot be told from one without. The answer has to be per paper, since ? is punctuation in every other paper here, and it is not applied.

Page 6 of that paper is a comparative appendix in Upper Chehalis and Cowlitz, thirteen sentences of two other Salishan languages. The table names each of them in its own column, which retains them out of a Lushootseed corpus.

8.4.2 *The damage is a mutation distribution*

The damage is a hazard to route around, and it is also the one calibrated mutation sample in the archive. The damaged text is retained for that reason.

Two encodings of one text, aligned line for line, with the page settling which is right, is a measured channel. The extracted text is the mutated side, the drafted page text the other, and the hand extraction from the rendered page is ground truth for both. No other paper here has a second encoding of the same page to compare against.

The mutations sort into five types, and the last two are what make this a distribution instead of a rewrite table.

Type	Instances	Invertible
Substitution, one symbol for one	ə→@, ɬ→ì, ʕ→Q, ʔ→P, x̌→ň, ·→;	yes
Collapse, two symbols onto one	ʷ→w, and plain w→w	no
Transposition	a combining mark over its character becomes a spacing mark before it: ǩ→ʻk, x̌→ˇx	yes
Insertion	a space appears in front of a character carrying a mark	not decidable alone
Deletion	the space at a change of font is dropped, welding two words	not decidable alone

The insertion does not always fire. Lindley and Lyon's text holds wa 'y in max places and wa'y in three, for the single word the page prints as waẏ. Its rate lies strictly between 0 and 1, and the

same input shape has two outputs. The insertion fires with a probability, and a repair applied without the page welds words that were never one word.

The deletion fires where the paper changes font mid-sentence. Footnote 5 of the same paper sets its cited forms in italic inside roman prose, and the two arrive as `Okanaganyaxʷt` and `Okanaganyaʔyáʔt`, each one token where the page prints two words. The second is recoverable, because `yaʔyáʔt` is a common word and appears on its own elsewhere in the paper. The first is not: `yaxʷt` occurs nowhere else. Which of the two happens depends on the rest of the corpus and not on the token, the second reason this is a distribution.

The collapse sets a ceiling. No recovery separates page `kʷukʷ` from page `wist` in the mutated text, because the channel destroys the distinction instead of disguising it. Anything downstream that claims to have recovered a labialized consonant on these two papers is claiming something the file cannot support.

A detector keyed to a glyph inventory reads the mutated side as holding no language at all, and reports a paper with eleven unread texts as fully covered. A detector reading the distribution of symbols survives the same input. The gold standard corpus says how well a vector separates the target language from the English around it; this pair says how much mutation it takes before it stops separating them at all.

The per-symbol rates are not measured. A candidate mapping is scored by how many mutated tokens become forms attested in a clean paper, which is a proxy for the answer. The page-verified table is the answer itself for the lines it covers.

The conversion table is per document and not per font family. Four of the six papers above are set in NimbusRomNo9L, and the two with page-verified tables supply a candidate mapping for the other two. Neither takes it, for two different reasons. Abraham's paper has nothing to convert. Lyon's paper on the linguistic evidence for a Francis Drake landing in Oregon [21] is damaged and wants a different table: its language data comes from Frachtenberg's Oregon transcriptions, and while `ì` for `ɬ` carries over and so does the transposed acute, it also needs a transposed macron the Okanagan papers never use, `E` for `ɛ`, and several more marks. One code disagrees outright. `;` is the raised length dot in the Okanagan papers and the glottalization mark here. Page 24's `k'eu'ts!` arrives as `k;eu´ts!`, and the Okanagan table would silently turn every glottalization into a length mark. Scored, it puts 4 of 407 occurrences into attested forms against 5 before it.

Robertson's paper [31] is damaged four ways, and one of them costs nothing. Its extractor could not resolve a glyph and printed the glyph's name instead. The text holds `/uni0294oo /uni026C /xé/uni0294` where page 30 sets a morphemic line. A `/uniXXXX` name carries the code point it is for: `/uni0294` is `ʔ`, `/uni026C` is `ɬ`, `/uni0259` is `ə` and `/uni019B` is `ƛ`. All 103 of them are turned back, along with `/combiningdotbelow` and `/combiningacuteaccent`, and no glyph name is left located anywhere in the paper. That part is arithmetic and exact. The other three come off the pages:

- **Labialization, collapsed exactly as it is in the Lyon papers.** Page 30 sets `/kʷú[·kʷ]piʔ` and the text gives `/kwú[·kw]pi/uni0294`. The paper's own prose loses it too: page 30 names its symbols `(č, š, xʷ)` and the text holds `( č, š, x w)`, with a space where the raised character was.

- **A word broken across two lines with no hyphen.** 42 lines of the extraction hold nothing but a fragment: `Da` above `vid D. Robertson`, `ma` above `ximal efficiency`. The majority of

them are in the interlinear, where the break falls inside a gloss: `A` above `UG-`, `s` above `ick`, `gr` above `eet`. A rule that joins every fragment line to the one under it would be wrong in the alphabet tables, where `wa`, `wi` and `waw` are Chinuk pipa character names and each is a line of its own.

- **A glyph repeated where the page prints it once.** Page 1's epigraph prints `t'íxʷəɬ ɣ-ʔ-[χ-]q'y-n'-tén` with one `ɬ` and one `ʔ`, and the text holds four of each. The doubling happens on 2 lines in the paper and both are set in bold italic.

The hand extraction is 195 rows and covers both Salish texts whole, the Chinuk pipa and Chinook Jargon forms cited through the analysis, and the two bibliography entries written in Chinuk Wawa. Texts 3 through 6 are Chinook Jargon, which is a pidgin; they are the paper's own material and they are not Salish, and the speaker column retains them out of the target corpus.

The extractor finds all 20 stanzas of Text 1 and all 7 of Text 2 with their numbering intact. The transliteration row is plain ASCII: `o l ha l kukpi,` carries no character of the modern orthography, and the test every other extractor leans on would call it English. The row is identified by where it sits. The gold standard corpus takes the transliteration row alone: the morphemic row is Robertson's analysis and holds forms nobody said. The line join is applied identically by the extractor and the coverage check, and with it the paper is at 100 percent coverage.

8.4.3 *The per-run test reads the damage a blanket repair cannot*

Reading a broken word off a rendered page works and does not scale: twenty papers at twenty-four pages each is four hundred images, and every reading is a judgment nobody can check afterward. The screening can be asked instead.

A break site is a place where the extraction has two tokens and the paper may have had one. The candidates are the two joined by each combining mark the paper writes, and joined by nothing, the reading where the space is a real boundary. A candidate counts as attested where the paper writes it as a token, and where the paper writes it inside one: `qʷal út` is `qʷaɬút`, which Davis and Mellesmoen [4] never print alone and do print inside `qʷəqʷaɬút`. Scoring is the per-run test of the dialect border, with run counts taken from the tokens carrying no break and flattened to max entropy. A run common everywhere contributes nothing, and what is left belongs to that restoration.

It finds 86 sites across the fifteen papers with extractors, and every one of those papers reports 100 percent coverage. That is not a contradiction: the coverage check puts both sides through the same repair, and a word broken in the source and broken in the extraction matches itself.

paper	read	declined on score
Garcia, Hannon and Stacey	24	9
Hall and Phillips	24	0
LaFontaine and Janzen	18	2
Davis and Mellesmoen	4	1
Kim	4	0
Mary George	4	0
Alexander and Davis	2	2
Nater, ICSNL 59	2	0
Lyon, inchoatives	2	0
Matthewson and Redan	1	1
Wolfe	1	0

`Kʷ əɬtəzétkʷu` reads as `Kʷəɬtəzétkʷu` at a score of 478.0, and that is Kʷəɬtəzétkʷu's own name broken in half in a paper that passes every check. `N ɬe?kepmxcín` reads as `Nɬe?kepmxcín` at 302.8 and `s x̌e?xímn.` as `sx̌e?xímn.` at 244.1. A mark drawn at random from that paper's inventory lands on an attested form 0.000 of the time over 6600 draws.

The blanket repair cannot reach any of them and is right not to. The break follows `ʷ`, which is outside the joining set because it is a spacing modifier character on the baseline and a space after one is normally a real boundary: including it welds thirteen pairs of real words in one paper. A blanket rule has to choose once for the whole corpus. This decides per site, from what the paper itself writes. It can take `Kʷ əɬtəzétkʷu` and leave `bəlkʷ 'return'` alone.

The score separates them. Garcia's `w ɬ` scores −1.4 and is declined, because the runs `wɬ` would make sit below where a flat distribution puts them: the paper does not write words shaped that way. Nine of Garcia's 33 sites are declined on that test.

These are candidate corrections, and none is applied. Applying them rewrites the corpora and moves what every downstream measurement reads, which is a decision to take deliberately and measure. The table establishes that the damage is there, that it is larger than the checks can see, and that it is readable without a person opening a page.

8.5 WHAT THE GOLD STANDARD CORPUS MEASURES

The gold standard corpus that the language profiles are built from is 2314 lines and 16898 tokens across the eleven papers that have both an extractor and a profile to count under. 749 of those lines and 6762 of those tokens, 40 percent of it, come from the two papers set in the font that renumbers its codes.

Robertson's paper writes gold standard lines of its own and is not in that total: it is Thompson and Secwepemctsín written in Chinuk pipa, and there is no profile for it. Wolfe holds eighteen languages and writes no single-language file at all. The two Nater word lists, the Lyon inchoatives and the Kim reduplications each write one, and none of the four joins a profile, because a word list and a narrative in one language sit about as far apart as two languages do: Nater's voiceless list measures 0.7783 from the Nuxalk narrative corpus and Lyon's inchoatives 0.6436 from Nsyilxcən's, against between-language distances of 0.51 to 0.83. Joining one to a profile would move the profile toward the shape of a dictionary page.

The extractors for the two damaged papers read the drafted page text. Reading the extraction instead would put the font's alphabet in the corpus, with only the substitutions the font repair covers applied to it. Lindley and Lyon, whose table is transcribed from the pages in full and takes no repair, measures the difference token for token:

	reading the extraction	reading the page text
corpus tokens the table holds	1231 of 1878, 66%	2204 of 2535, 87%
marks located in front of their character	1832	54
labialized consonants written with a plain `w`	1630	42

The last two rows are for both Lyon papers together. Read from the extraction the corpus holds `iks'ma'yɬtím` where the page holds `iksm̓aẏɬtím`, and `sqwlqwlstwíxwtət` where the page holds `sqʷlqʷlstwíxʷtət`. Read from the page text it holds what the page holds.

What is left of the 13 percent is the inserted space: a word arrives in pieces and the corpus retains the pieces. `incítxʷ` is in it as `incítx w`. The space repair cannot reach these, because the

vocabulary it joins with comes from the interlinear's form column, which in this paper is the parse and carries morpheme hyphens the running text does not.

Every repair in the page-text path returns its line unchanged, because applying the mapping to text that has already been through it destroys the text: `P` to `?` turns `Papers` into `?apers` and the gloss label `APPL` into `A??L`. The language test on that path asks about the orthography itself, since the page text holds none of the damaged characters by construction.

8.6 WHAT A GOLD STANDARD CORPUS IS FOR

The purity of both sources is the constraint everything else answers to. The target is the Salishan corpus and the reference corpus is English, and a boundary drawn between two distributions says nothing about the languages if either side is holding something other than what it claims.

A paper holding eighteen languages has no single gold standard stream. Every other extractor writes the lines of one language, because every other paper is one speaker telling one story, or one language under analysis. Wolfe cites lexical suffixes from published dictionaries of Sliammon, Sechelt, Squamish, three types of Halkomelem, three types of Straits, Klallam, Lushootseed, Twana, Tillamook, Quinault and two Tsamosan languages. Pouring those into one file builds a corpus of no language at all, and it would be a corpus the screening is then measured against, where that mistake cannot be recovered from. Its extractor writes its gold standard lines keyed by language, and the paper joins no profile.

Two further things are excluded. A reconstruction is a form the author worked out and nobody has ever said, the same distinction as between a transcription and a segmentation line, applied to a proto-form. And the predicted reflexes of Tables 6 and 8, which give for each language the form expected if the reconstruction was stressed, the form expected if it was not, and the form attested; the extraction does not retain those three columns apart reliably. Nothing is lost, because the attested forms of both tables are printed again in examples (1) and (6) where the columns are unambiguous.

The hand extractions can also be turned on themselves. Each is verified against its own paper. They should agree with each other about what these languages look like, and one can be measured against the rest taken together as the reference corpus. A row that sits oddly against everything else verified is one of two things: a slip in the transcription, or a real property of that paper nobody has noticed. Both are worth opening the paper for, and neither is visible from inside one table. At the language level this runs: each dialect is measured against the other five pooled, the question the screening is actually asked, and all six are off the pool. Per row it does not run, and per row it would name a paper to open.

Once each dialect's corpus is pure, the dialects become each other's reference corpora. A boundary drawn between Nsyilxcən and Nɬeʔkepmxcín is a claim about where one ends and the other begins, and it is worth drawing because of what sits near it: forms that two dialects share and a third has lost, which is where onomatopoeia that has gone out of one of them would show.

One such boundary is drawn and scored against an answer the algorithm did not supply. Lushootseed is not one dialect, and Mellesmoen and Kye's stress paper [25] labels every form it cites northern or southern. The hand extraction carries that label in the speaker column, a published fact from one paper, in one orthography, by one transcriber. The border test loads those labels, sets them aside, and only compares at the end. Comparing whole distributions fails at this

size and says so: the two varieties hold 1469 and 2768 bytes against a method that resolves at 6707, and a blind partition scores exactly the majority class. Holding the concept fixed with the word web's gloss edge, flattening the pooled run counts to max entropy, and testing one run at a time does find it. The term is the stressed schwa ə́, southern, 45 against 6 over the 58 concepts both varieties name. Putting the border back at random 200 times, 1 of those random borders finds as much. The chapter *How wrong it could be* carries the tables.

That comparison has to run on a sound representation and not on the characters. These orthographies write the same sound differently and the difference is arbitrary. The lateral fricative is ɬ in one paper here and ƛ in another. The glottal stop is ʔ in max and the digit 7 in the van Eijk papers. Labialization is a raised ʷ in some and a plain w in others. A model counting character bigrams reads two dialects that share a word as maximally far apart, because it is measuring the transcriber and not the speaker. Encoding each segment as its features and comparing those makes the distance a fact about the languages.

The plan runs in three stages. First the gold standard text corpus, which fixes what each dialect writes and which the hand extraction workflow exists to produce. Then a sound representation measured off recordings, where a segment is a set of binary features and the text space maps onto those features one written symbol at a time. Then a probability distribution over that representation, the thing an interdialect boundary is drawn on, with the word webs built on the encoded forms so that two dialects writing one word two ways arrive at one node.

The recordings the sound representation is measured from are three, all nɬeʔkepmxcín, all Bev Phillips. That is enough to build an encoding against and short of what a second dialect would take.

The word web is built on the written forms. It joins every form in the hand extractions to the forms it is related to, by three types of edge. A **concept** edge joins two forms whose glosses share a content word; it crosses an orthography, because the gloss is the only part of a form two papers wrote the same way. A **shape** edge joins two forms sharing a leading or trailing run of four characters, which in these languages is usually a shared root or affix. A **context** edge joins two forms written in the same section of one paper by one speaker. The web is at 7062 forms and 373575 edges across nineteen groups. A dictionary multiplies the edges of a web keyed on glosses: 1275 entries each carrying an English gloss give the concept edge far more to join than the same number of words in running narrative.

The join between the web and the sound representation is not built: nothing takes a character out of the text space and passes back the features it is for. The web's nodes are written forms, and two dialects writing one word two ways are two nodes. The text space is that join's other side, and it stays one definition for that reason.

The screening's calibration reports the separation as healthy against that corpus. Under one profile, 2285 known gold standard lines sit at a median of 15.15 against 1899 known English spans at 8.49, and a cut at 9.28 retains 99.0 percent of the gold standard corpus while admitting 15.3 percent of the English. Under two profiles, 96 percent of the gold standard lines come back as language and 98 percent of the English spans as English. Those figures say the two profiles are far apart. They do not say the language profile is the language; the chapter *The corpus under the instrument* gives what the six per-language profiles do, where they fail to separate, and why. The same measurement on the other papers is not comparable, because each goes through its own

repair before the comparison, and several of those orthographies write labialization with a plain w.

8.7 THE BINARY SOUND REPRESENTATION

The representation turns a recording into two bit fields per 10 ms frame. Four facts about hearing set its shape.

The waveform on its own is not enough. A vowel is a harmonic series under an envelope, and two recordings of one vowel share the envelope while the harmonics sit wherever the speaker's pitch put them. The analysis is spectral and the envelope is smoothed off the source before anything is compared.

What a person hears is not what the microphone recorded, and it differs between listeners by physiology and by how much hearing they have lost. The weighting takes an audiogram, the six frequencies a hearing test is run at with the loss in dB at each; the loss comes off a band before loudness, and a band a listener cannot hear stops reaching the code. The default is a listener who has lost none.

Some sounds are the same sound at different frequencies. A phone said by a small speaker and by a large one differs by a scaling of the whole spectrum. The 64 bands are log spaced, and that scaling is then a shift along the band axis, and the magnitude of a Fourier transform along that axis is unchanged by the shift. That transform is the segment field.

And some sounds that differ only in frequency mean different things. Stress carries an acute in every orthography in this corpus, and a field made shift invariant would put a stressed form and an unstressed one at one address. The prosody field is separate for that reason and holds loudness, pitch against the speaker's own median, how that pitch is moving, and voicing.

Bringing the sample to max entropy makes the delta readable. Each bit is cut at its own column's median over the recording and splits the frames in half, and the segment columns are rotated onto their principal axes first so that no bit restates another. All 24 segment bits come out set on 0.500 of frames on all three recordings, and all 64 of the six bit states and all 256 of the eight bit states are reached on all three. A recording that used those states evenly would be noise, and the distance from that flat reference is the structure in it.

Recording	Frames	8 bits	10 bits	12 bits	Prosody
Givens and Hall, ICSNL 58	23489	0.2157	0.2538	0.3161	0.3708
Hall and Phillips, ICSNL 59	83249	0.2364	0.2863	0.3207	0.3498
Hall and Phillips, ICSNL 60	134997	0.1859	0.2200	0.2519	0.4594

Each number is the total variation between the codes that width of field took and the flat distribution over its states. The width is part of the figure because the field has to be sampled to be read. At 12 bits the longest recording puts 33 frames in each of 4096 states; at 24 bits it puts 133820 codes in 16.8 million states, which measures the frame count and not the recording.

All three recordings came from the nłe?kepmxcín Lab and all three are Bev Phillips. These numbers are the method working and not a comparison of dialects. A second dialect's recording turns them into a reading.

8.8 AUDIO SOURCES

Recording	Size
Hall and Phillips, xʷiʔ kʷ páq (You Will Be Sorry), ICSNL 59 [8]	13.4 MB
Givens and Hall, The Moon and the Birchbark Canoe, ICSNL 58 [7]	7.5 MB
Hall and Phillips, ɬ cutés us ɬ qəɬmín ɬ tmíxʷ, ICSNL 60 [9]	17.1 MB

The third is Bev Phillips reading the story whose paper has an extractor, and it is a test oracle for that extraction as well as a source.

Recordings named in papers and not published with them:

- Qwa7yán'ak, Graveyard Valley narrative. Just over half an hour, recorded by Henry Davis at Nxwísten on 7 July 2025.
- Ḵweswapáẇ (Linda Redan). Three minutes twenty-eight seconds, told over Zoom on 31 October 2025. Audio and video are retained by her.
- George Lezard, 1966, recorded by Randy Bouchard.
- Nellie Guitterez, 1978 or 1979, recorded by Yvonne Hébert at an Elders' Gathering in Quilchena.

8.9 UNREAD PAPERS

152 extracted paper texts are stored, and 146 of the PDFs they came from. Eighteen have an extractor and a hand extraction, and seven more have a hand extraction alone. Six of the 146 are in the font class above, and their texts are the font's encoding. Fifty of the papers are about Lushootseed or Skagit.

The screening reads each paper's front matter for the names the literature uses for each language and attributes a paper where it names one and no other. Over 154 texts it attributes 122 and leaves 32, which are the comparative papers and are correctly left. Lushootseed leads at 43 papers, Nuxalk follows at 32.

Counting how much of a language each unread paper actually holds gives a different order, and it is the order to read by. Lushootseed has 40 unread papers and 593 distinct language tokens across all of them together, which is less than one Nater appendix. The majority of the Lushootseed literature is older and writes the glottal stop as `?` and the schwa as `a` or `E`, and a mark set built on the modern orthography reads those papers as holding no language at all. Comox has 8 unread papers and 1899 tokens, St'át'imcets 10 and 1297, Nuxalk 29 and 1046.

The count of papers per language is not the count of language per language. What Lushootseed needs is not another paper but the treatment Hilbert's 1983 paper and Hilbert and Hess already have: a per-paper mark set for the orthography that paper is actually written in.

8.10 VERIFICATION SOURCES

Some sources test a transformation instead of supplying text. Each answers a question the code that made the change cannot answer about itself.

- Lyon's inchoatives paper [23], a later paper on Nsyilxcən whose extraction retained its characters. It tests the font substitution table, which moves attested tokens from 1 of 3599

to 811 on one paper and from 2 of 4332 to 965 on the other, and the word list recovered from the interlinear, where none of the broken forms has its pieces attested apart and 4 percent and 36 percent are attested once joined.

- Each paper's own second printing. Hall and Phillips, Garcia, and both Lyon papers print their stories twice, once as running text and once word by word. One printing checks a reading of the other.

8.11 COMMUNITY SOURCES

- Puyallup Tribal Language Program, `https://www.puyalluptriballanguage.org/`
- The Salish Institute, `https://www.thesalishinstitute.com/salish-language`

8.12 ARCHIVES

- ICSNL volumes, 1967 to present, every paper on one page: `https://lingpapers.sites.olt.ubc.ca/icsnl`
- UBCWPL, volumes 33 (1998) to 61 (2026): `https://lingpapers.sites.olt.ubc.ca/`.
- ICSNL archives, volumes 1 to 32, the Kinkade Collection: `https://blogs.ubc.ca/icsnl/icsnl-archives-a`
- nɬeʔkepmxcín Lab, the source of the three recordings: `https://blogs.ubc.ca/nlab/projects-publicatio`
- ICSNL 50 as one volume: `https://lingpapers.sites.olt.ubc.ca/files/2018/01/ICSNL2015-fullonlin`

BIBLIOGRAPHY

[1] Marie Amhatsin' Abraham. *A Sasquatch story in St'át'imc.* Papers for the International Conference on Salish and Neighbouring Languages 50, UBCWPL 40, https://lingpapers.sites.olt.ubc.ca/files/2018/01/21-Abraham_ICSNL50_final-4.pdf. 2015.

[2] Carl Alexander (Qwa7yán'ak), Henry Davis. *I Tsícwas sQwa7yán'ak Áku7 Graveyard Valley ('When Qwa7yán'ak went to Graveyard Valley'): A St'át'imcets Narrative.* Proceedings of ICSNL 61, UBCWPL, https://lingpapers.sites.olt.ubc.ca/files/2026/07/AlexanderDavis_ICSNL61.pdf.

[3] W. B. Cavnar, J. M. Trenkle. *N-Gram-Based Text Categorization.* Proceedings of the Third Annual Symposium on Document Analysis and Information Retrieval, 161-175. 1994.

[4] Henry Davis, Gloria Mellesmoen. *A Newly Discovered Reduplication Pattern in St'át'imcets and its Implications.* Papers for the International Conference on Salish and Neighbouring Languages 58, UBCWPL, https://lingpapers.sites.olt.ubc.ca/files/2023/08/ICSNL58_Davis_Mellesmoen_final.pdf.

[5] William W. Elmendorf. *Word Tabu and Change Rates: Tests of a Hypothesis.* Second International Conference on Salish Languages, University of Washington, Seattle, August 1967.

[6] Bernice Garcia, Ella Hannon, Anna Stacey. *Three Glossed Nɬeʔkepmxcín Narratives by Kʷəɬtəzétkʷu (Bernice Garcia).* Papers for the International Conference on Salish and Neighbouring Languages 59, UBCWPL, https://lingpapers.sites.olt.ubc.ca/files/2024/07/ICSNL59_Garcia_Hannon_Stacey_final.pdf.

[7] Katherine Givens, Brent Hall. *The Moon and the Birchbark Canoe (ɬ máʕxetn pe ɬ qʷɬinéwɬ).* Papers for the International Conference on Salish and Neighbouring Languages 58, UBCWPL, https://lingpapers.sites.olt.ubc.ca/files/2023/07/ICSNL58_Givens_Hall_final.pdf; recording https://lingpapers.sites.olt.ubc.ca/files/2023/07/ICSNL58_Givens_Hall_The_Moon_and_the_Birchbark_Canoe.mp3.

[8] Brent Hall, Bev Phillips. *xʷíʔ kʷ páq (You Will Be Sorry).* Papers for the International Conference on Salish and Neighbouring Languages 59, UBCWPL, https://lingpapers.sites.olt.ubc.ca/files/2024/07/ICSNL59_Hall__Phillips_final.pdf; recording https://lingpapers.sites.olt.ubc.ca/files/2024/07/ICSNL59_Hall_Phillips_audio.mp3.

[9] Brent Hall, Bev Phillips. *ɬ cutés us ɬ qəɬmín ɬ tmíxʷ (When Old One Created the Earth).* Papers for the International Conference on Salish and Neighbouring Languages 60, UBCWPL, https://lingpapers.sites.olt.ubc.ca/files/2025/07/HallPhillipsICSNL60.pdf; recording https://lingpapers.sites.olt.ubc.ca/files/2025/07/2025-05-29_BP_storyrevised_trimmed.mp3.

[10] Brent Hall, Noah Luntzlara, Gloria Mellesmoen, Danica Reid. *Ctrl-Alt-Delete: The Control Directive and Associated /t/ Deletion in nɬeʔkepmxcín*. Proceedings of ICSNL 61, UBCWPL, https://lingpapers.sites.olt.ubc.ca/files/2026/07/Hall-et-al.-ICSNL_61-1.pdf.

[11] Eric P. Hamp. *Another Look at Tillamook Phonology*. Second International Conference on Salish Languages, Seattle, August 1967.

[12] Thom Hess. *The Morph /-(ə)b/ in Snohomish*. Second International Conference on Salish Languages, Seattle, August 1967.

[13] Vi taqʷšəblu Hilbert. *Poking Fun in Lushootseed*. ICSNL 1983, https://lingpapers.sites.olt.ubc.ca/files/2018/03/1983_Hilbert.pdf.

[14] Vi Hilbert (taqʷšəblu), Thom Hess. *A Note on ʔə Constructions in Lushootseed*. https://lingpapers.sites.olt.ubc.ca/files/2018/03/1975_Hilbert_Hess.pdf.

[15] John Hamilton Davis. *Mary George Personal Narratives*. Papers for the International Conference on Salish and Neighboring Languages 56, UBCWPL, https://lingpapers.sites.olt.ubc.ca/files/2021/08/ICSNL56_DavisJ_2_final-1.pdf.

[16] Hyung-Soo Kim. *The truncated reduplication in Twana: Another case of synergistic weakening*. Papers for the International Conference on Salish and Neighbouring Languages 52, UBCWPL 45, https://lingpapers.sites.olt.ubc.ca/files/2017/07/Kim_TwanaReduplication_final.pdf.

[17] M. Dale Kinkade. *Deictics in Columbian: A Work Paper*. Second International Conference on Salish Languages, Seattle, August 1967.

[18] Jade LaFontaine, Jonathan Janzen. *Four Stories by wlwlmelst*. Papers for the International Conference on Salish and Neighbouring Languages 59, UBCWPL, https://lingpapers.sites.olt.ubc.ca/files/2024/07/ICSNL59_LaFontaine_Janzen_final.pdf.

[19] Lottie Lindley, John Lyon. *12 more Upper Nicola Okanagan narratives*. https://lingpapers.sites.olt.ubc.ca/files/2018/01/2013_Lindley_Lyon.pdf.

[20] Deryle Lonsdale, Hitokazu Matsushita. *Annotating and exploring Lushootseed morphosyntax*. https://lingpapers.sites.olt.ubc.ca/files/2018/01/2011_Lonsdale_Matsushita.pdf.

[21] John Lyon. *Francis Drake's 1579 voyage: Assessing linguistic evidence for an Oregon landing*. Papers for the International Conference on Salish and Neighbouring Languages 49, https://lingpapers.sites.olt.ubc.ca/files/2018/01/Lyon-final.pdf. 2014.

[22] John Lyon. *Three Okanagan stories about priests*. Papers for the International Conference on Salish and Neighbouring Languages 50, UBCWPL 40, https://lingpapers.sites.olt.ubc.ca/files/2018/01/19-Lyon_ICSNL50_final-78.pdf. 2015.

[23] John Lyon. *Nsyilxcn Inchoatives and their Distributions Across Root Types*. Papers for the International Conference on Salish and Neighbouring Languages 60, UBCWPL, https://lingpapers.sites.olt.ubc.ca/files/2025/07/LyonICSNL60_Inch-2.pdf.

[24] Lisa Matthewson, Ḵweswapáẇ/Linda Redan. *Cw7aoz káti7 láti7 ku naxwít: A story by Ḵweswapáẇ/Linda Redan*. Proceedings of ICSNL 61, UBCWPL, https://lingpapers.sites.olt.ubc.ca/files/2026/07/Matthewson_Redan_ICSNL61.pdf.

[25] Gloria Mellesmoen, Ted Kye. *A Comparative Analysis of Stress in Northern and Southern Lushootseed*. Proceedings of ICSNL 61, UBCWPL, https://lingpapers.sites.olt.ubc.ca/files/2026/07/Mellesmoen_Kye_ICSNL61.pdf.

[26] Hank Nater. *How Salish is Bella Coola?*. https://lingpapers.sites.olt.ubc.ca/files/2018/01/2013_Nater.pdf.

[27] Hank Nater. *A Bella Coola tale*. Papers for the International Conference on Salish and Neighbouring Languages 50, UBCWPL 40, https://lingpapers.sites.olt.ubc.ca/files/2018/01/22-Nater-Bella-Coola-tale-10.pdf. 2015.

[28] Hank Nater. *Voiceless Words in Bella Coola: Fact vs. Fiction*. Papers for the International Conference on Salish and Neighbouring Languages 59, UBCWPL, https://lingpapers.sites.olt.ubc.ca/files/2024/07/ICSNL59_Nater_2_final.pdf.

[29] D. Quigg. *Delta Null*. orior research papers, https://github.com/dstroy0/orior. 2026.

[30] D. Quigg. *Workbook*. orior research papers, https://github.com/dstroy0/orior. 2026.

[31] David D. Robertson. *BC Indigenous people's Chinuk pipa script: History, analysis, and texts*. https://lingpapers.sites.olt.ubc.ca/files/2018/01/2012_Robertson.pdf.

[32] C. E. Shannon. *A Mathematical Theory of Communication*. Bell System Technical Journal 27, 379-423 and 623-656. 1948.

[33] Julie Wolfe. *Lexical Suffixes and Connectives in Proto-Central Salish and Beyond*. Papers for the International Conference on Salish and Neighbouring Languages 60, UBCWPL, https://lingpapers.sites.olt.ubc.ca/files/2025/07/WolfeICSNL60.pdf.